Every Bag in Its Place

A novel of the Luneburg Trail Run, with the screens that prove it

Two hundred and sixty-two goody bags, four races, one weekend, and every way it can go wrong. Names, entries and race numbers are invented.

Every picture in this book is a real screen. None of them is a drawing of what the system might look like. Each one was photographed by a test that plays the whole weekend against the running software, on an entry list built to contain every exception in the story: a registration covering three people in two classes, an entrant with no race number, an entrant with no email address, an unpaid entry, a race number issued twice, a family whose bags had to go without anybody's phone, two children under sixteen and a third whose father named him, a twelve-year-old with a school email address, and six replies to a letter. What the screens say is what the software said. Appendix C lists every case, what was sent to the desk, and what came back.

This is the third edition. The second took the PIN off the phones, stopped the camera acting on its own, gave the laptop a name on the Wi-Fi and the desk a third button. This one adds a rule about who may collect a family's bags, a last letter before the race that lets each family change it, a program that reads their replies and leaves alone the ones it cannot read, a screen for the ones a person has to settle, and a sheet of paper for the exception desk that nobody has to remember to print. The story has moved to fit, and every screen was taken again.

Part One: What Is On The Table

1. Thirty Columns

The EntryNinja export for the Luneburg Trail Run arrives in August with seventy columns and two hundred and sixty-seven rows, and about fifty of those columns are nobody's business.

Identity numbers. Home addresses. Medical aid numbers, blood types, allergies. A man from Vryheid who has written out, with touching thoroughness, the name of every tablet he takes. The desk does not need any of it. The desk needs a name, a distance, a number, whether they have paid, and an emergency contact for the medics.

So the first thing that happens to the file is that it is cut down. Seventy columns become twenty-seven, and the run says out loud what it threw away:

```
Wrote input/TestExport2026_slim.xlsx
  rows    14 -> 11 (rows in surname order)
  columns 70 -> 27
  size    9 kB -> 7 kB
  removed: home address, medical aid, medical aid number, medical information, national ID number,
telephone number, third party contact detail
  collection both: LTR 10, LTR 5, LTR 65, LTR 23
  letters internal: LTR 10, LTR 5, LTR 65, LTR 23
  positions assigned and printed for: no class (add :show to a --class line)
  no position anywhere for: LTR 10, LTR 5, LTR 65, LTR 23

Still personal data: names, emails, dates of birth, bag positions and
emergency contact numbers. Look after the file and the USB stick.
```

That last paragraph is not decoration. The identity numbers and the tablets stay in EntryNinja, where they belong, and the thing that travels to the marquee on a USB stick on a red lanyard is a third the size and holds a third the damage. What is left is still enough to ruin somebody's year if it is lost, and the program says so every single time rather than letting anybody forget.

What comes out is called *race_day.xlsx*, and it lives in *input*, beside the export it came from. Everything else the system makes — labels, letters, invitations, block maps, slips, the check-in record, the files for the timekeeper — goes into *output*, and nothing goes anywhere else. That division is not tidiness. It is so that on the Sunday, when somebody asks where the labels are, the answer is one word.

Anke Wiese, who built all of this over three weekends and now maintains it the way other people maintain a bakkie, has one rule pinned to her fridge:

> ***A stopped run is better than a shortened list.***

Every step accounts for every record. If two hundred and sixty-seven rows go in and two hundred and sixty-two come out, the five are named — member number, race number, name, class and reason — on the screen and on an *Excluded* sheet in the workbook:

```
14 records in the export, 11 kept, 3 excluded:
      1  not fully paid
      1  no race number
      1  duplicate race number

Excluded records (3), also written to the 'Excluded' sheet:
     member  race no  name                          class    reason
   21700210       31  Venter, Marius                LTR 65   not fully paid (No)
   21700211        -  Nel, Christo                  LTR 23   no race number
   21700213      133  Fourie, Lerato                LTR 23   race number already used by Fourie, Lerato
```

A list that quietly comes out five short is more dangerous than one that refuses to come out at all, because the labels print, the emails go, and the first anyone knows of it is a runner on race morning with nothing waiting for them.

2. One Rule for the Whole Marquee

Four races. The 65 km goes round the mountain and takes the fast ones ten hours. The 23 km, the 10 km and the 5 km are done by mid-morning, and eleven of the twenty-three entrants in the 5 km are Krugers.

For years the 65 km was scanned and the other three were not. The short races ran on a runner saying their race number and then their name, a volunteer finding the bag by its number and checking the label against the name, and a sticky note left showing where the bag had been. It was fast, it was warm, and it worked while nearly everyone in the queue was somebody a volunteer already knew.

Then Luneburg got known. Entries from Durban, from Gauteng, from a running club in Mbombela that brought a bus. By the third August the committee counted the addresses on the export and found that two out of every three entrants in the 23 km and the 10 km were people nobody at the desk had ever met — and a name means nothing when the face means nothing.

The arithmetic is in Appendix B, and it is not flattering to either side. Scanning never buys speed; it only buys correctness, and it costs roughly ten seconds a runner and some of the warmth of the thing. But what decided it in the end was not the crossover number. It was the thought of running two different desks side by side in one marquee, with two different rules, and volunteers walking from one to the other at seven in the morning.

So this year there is one rule for the whole marquee, on all four races, and it fits on a card taped to the table:

> ***Scan the runner. Then scan the bag.***

Every class. Every handover. Nothing else completes at the table.

The number-and-name conversation has not disappeared. It has moved. It is now what happens at the exception desk, at the far end of the marquee, when somebody cannot show a code — and the thing it produces is a piece of paper, which is the subject of Part Three.

The bags lie in race number order along one aisle, two tables deep, volunteers in the middle. The tens run along the tables; the units count across them, 0 at the front edge nearest the runners, 4 at the aisle, 5 just behind the volunteer, 9 at the back edge. Bag 247 is four columns along and seven across. The number is the position, which means nothing has to be printed on the bag to say where it lives.

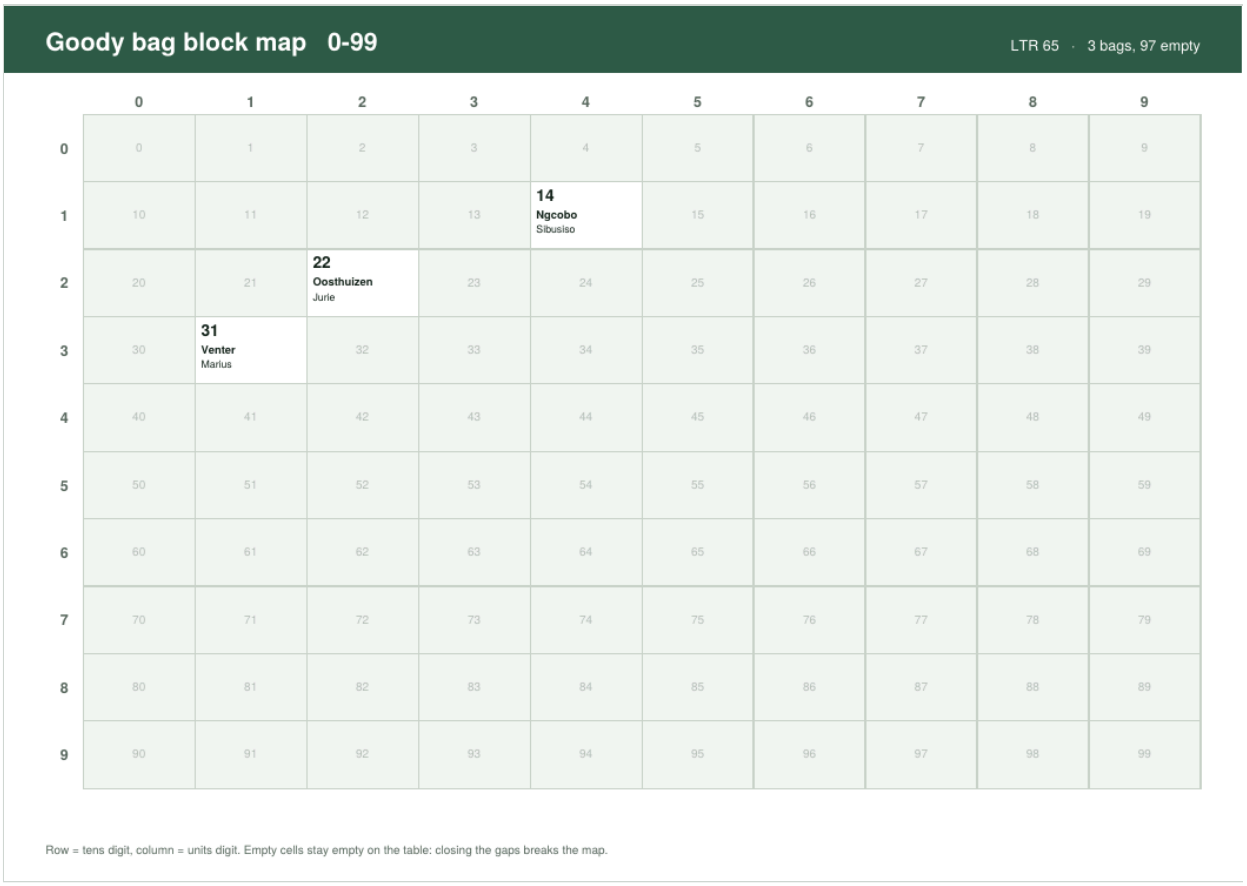


Figure 1. *The block map, one sheet per hundred, taped to the table legs. The number is the address: nothing on the bag says where it lives, so nothing on the bag can be wrong about it.*

What does go on the bag is a label, and there is one per goody bag: a code on the left, the name in the largest type that fits, the race number under it.



Figure 2. *A4 self-adhesive sheets, two across and six down. The labels now fill the sheet: a 2 mm border all round and 2 mm between them, which makes each one 102 by 47.2 mm. The code is the bag's own, and it is deliberately not the runner's: scanning a runner's code twice can never complete a handover.*

The sheets changed this year, and so did the printout. The old ones had a wide margin at the top and bottom and a gap down the middle, and the labels sat in the middle of the page like a photograph in a mount. The new ones are cut almost to the edge. Anke prints one calibration page first, holds it against a sheet of labels at the kitchen window, and the lines fall on the cuts.

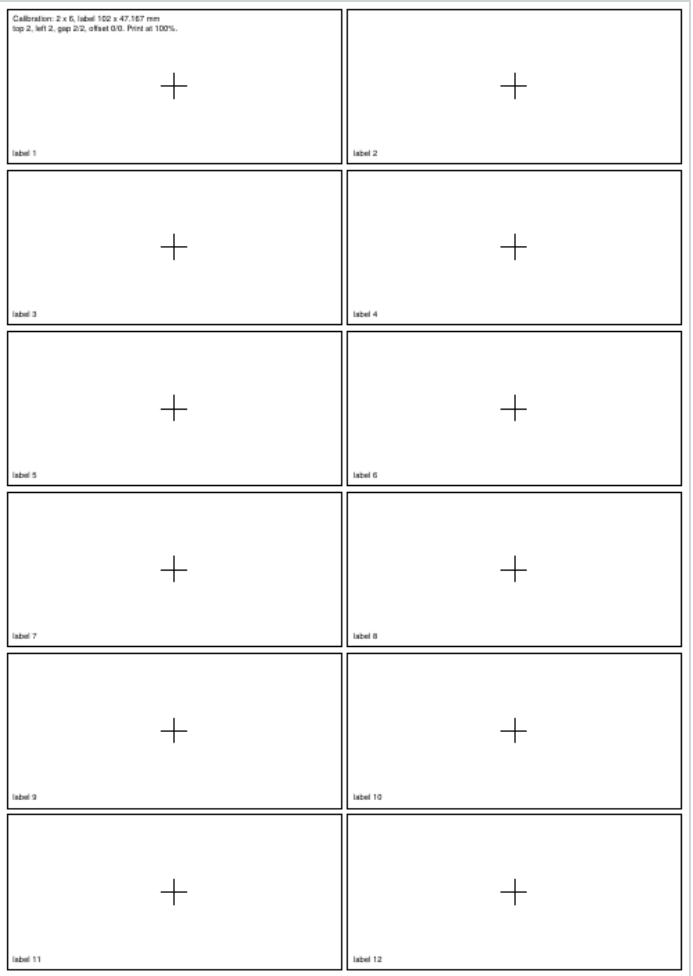


Figure 3. *One page of outlines before any label is printed. Twelve boxes, a cross in the middle of each, and the measurements written in the first box, so that a sheet that does not match says by how much.*

There is one rule about the bags on the table, written in capitals because it is the one that gets broken:

> ***NEVER CLOSE THE GAPS.***

A number nobody holds stays an empty square. The moment somebody tidies up by sliding the bags along, 247 is no longer four columns in, and the whole map is gone.

3. Two Phones and a Sheaf of Paper

When the check-in server starts, it prints two QR codes into the output folder, captioned, and those get printed and taped to the table.



Figure 4. *The first code opens the scanning app: the runner's code, then the bag. Every volunteer's phone at the table runs this one.*



Figure 5. *The second opens a different app entirely, with one green button on it, for the walk of the table at the end. It is deliberately not the same app.*

They are two apps rather than two screens in one app for a reason that took a season to learn. The walk of the table happens once, at the end, by somebody tired, and it must not be reachable by a thumb that meant to press next runner.

Look at the addresses under the codes. Last year they were a number, something like 192.168.1.23, which meant nothing to anybody and changed whenever the router felt like it. This year the laptop also has a name on the Wi-Fi — *lueneburg-trail-run* — and the story of that name is a short one with a lesson in it. Anke first gave it the kind of name printers use, *lueneburg-trail-run.local*, and it worked on her iPhone and on the laptop and on Johan's iPad, and on Thabo's Android phone it said *This site can't be reached: server IP address could not be found*. Android phones do not look up that kind of name, and two thirds of the volunteers carry one.

So the codes carry the number, which every phone can open, and the name is printed underneath for the phones that can. And on race weekend Thabo goes one step further: he tells the Wi-Fi router to ask the laptop for names, and starts the server so that it answers. Then every phone, Android or not, opens *https://lueneburg-trail-run:8443/*, the browser shows the race's name at the top of the screen, and a phone that has drifted onto somebody else's Wi-Fi finds nothing at all rather than something else. The number still works underneath it, because the name is a convenience and the number is the thing that always works.

And on the desk beside them is a sheaf of paper — five sheets of tear-off slips, one per class, plus a fifth for the registrations whose race numbers are spread across classes. Each slip carries a surname, the names, and every race number on that registration in figures you can read at arm's length. No QR code. Nothing to scan.

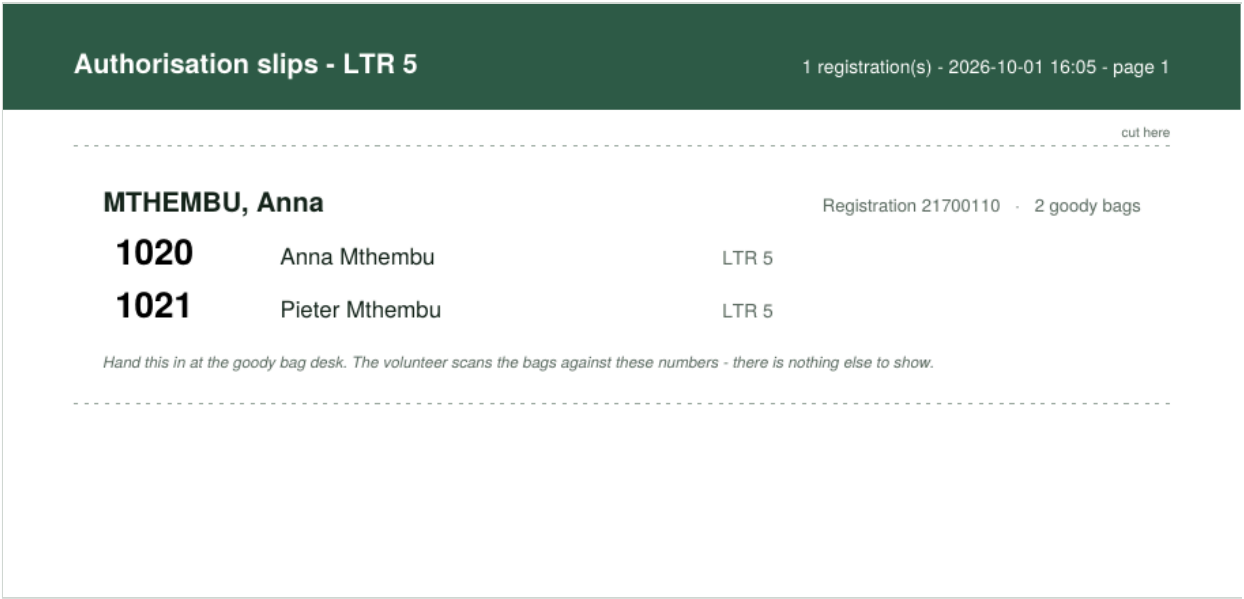


Figure 6. *One sheet of the five. Double-spaced, with a cut line above each slip, because they are meant to be torn.*

That paper is the whole of the exception process. When somebody cannot show a code — a flat phone, a letter that never arrived, a grandmother fetching four children's bags — the exception desk finds them, tears off their slip and hands it over. The slip is gone from the only sheet it was ever on, and the gap in the sheet is the record of what the desk gave away.

4. Twenty-Two Alerts

Anything that needs a person to decide something prints in red, between two rows of hashes, with a number:

```
##### ALERT 07 #####
1 record(s) left out: their race number is already used by someone else
  Fourie, Lerato (LTR 23)
Action: Fix the numbering in EntryNinja and re-run everything that prints a number.
See the setup guide, Alerts, number 07.
##### END ALERT 07 #####
```

There are twenty-two of them, two more than last year: one for a family's choice that no longer fits the entry list, and one for the replies a program could not read. The numbers are stable, because they are printed, quoted in the setup guide, and read out over a radio on race morning. Most say what was left out and leave you to decide whether that was right. Three stop everything: two people sharing one identity, one race number used twice in the column the system is using to tell people apart, and a class too big for the table it was given.

What follows is what happened to all of it.

Part Two: Before Anything Is Printed

5. Two Women, One Number

The first thing the system does, in the third week of August, is refuse to be clever.

Race number 133, in the 23 km, belongs to Lerato Fourie. It also belongs to Lerato Fourie. Her husband entered her as a surprise in June, not knowing she had entered herself in May, and EntryNinja took the money twice and issued the number twice.

Anke has seen this fail the other way at other races: the second row is dropped, quietly, because a spreadsheet can only hold one row per key. Lerato gets a label and an email and a bag, and the duplicate entry sits in the accounts unnoticed until February.

Here the run tells you before it does anything else. Alert 07 names her, the *Excluded* sheet carries the reason in words — *race number already used by Fourie, Lerato* — and the second entry gets no label, no letter and no square on the table. Johan phones Lerato on the Thursday; she laughs for some time and the duplicate is refunded the next morning.

There is a worse version of this, and the system has a separate pre-flight for it. Before anything is printed at all, *check_ids* reads the export and works out which column can safely stand for one entrant:

```
Recommendation:
  Use Member #. Pass it to every script as --id-col "Member #",
  or set id_column in config.ini so all of them agree.
  Also unique: Registration #+Member #, Order #+Member #
```

Had the system been keyed on Race Number, Lerato's twin entry would have collided with the very thing used to tell entrants apart, and the run would have stopped dead rather than excluded anybody:

```
STOPPED: 1 row(s) share a value in 'Race Number', which this
system uses to identify one entrant. Those people would be silently left out of
labels, emails, bibs and the check-in list, so nothing has been written.
```

Nothing written. Not a short list — no list.

6. The Eleven

On the Tuesday, Johan comes round with a folder of paper he does not need and a question he does.

"What happens to the ones who haven't paid?"

Anke turns the laptop round and presses *Non-compliant members*, and there they are, all eleven at once, with what is wrong beside each name: not fully paid, or no race number yet, or both. It takes him forty minutes, because he knows nine of them.

"This one. Card declined in July, phoned me twice, he's coming, he'll pay at the desk." The farmer from Paulpietersburg.

"This one entered twice and cancelled the second and the system's confused itself."

Then he stops at a name in the 23 km, takes his glasses off, and says: "Entered in May. Never paid, never answered. Three emails and a phone call. He's not coming, Anke. He entered on a Sunday night and thought better of it on the Monday."

Beside each name is a box with three words in it: *leave*, *terminate*, *nudge*.

Terminated, that man comes off the list. No bag, no square on the table, and on Saturday when Johan asks who is not here, the answer is a line on the Excluded sheet with a reason, not a hole in the middle of the 23 km.

The other eight are nudged, which does the opposite thing. A nudged entrant is imported whatever the rules say — unpaid, unnumbered, it does not matter — and gets one last letter that says what is missing and where to go.

> Your entry is not complete yet, so there is no goody bag waiting for you. Please come to the late entry and compliance desk at registration: the organisers will sort it out with you there and you can still run.

With an orange panel underneath saying what is outstanding: payment of your entry, or a race number, or both.

Johan reads it twice. "It doesn't tell them off."

"No."

"Good. Half of them are farmers in a bad year."

The decisions save into a small file in the input folder, so the next time anybody slims the export — Anke, or Thabo, or whoever comes after them — the same eleven people are treated the same way without anybody having to remember the conversation.

7. One Email, Three Bags

The Kruger registration is the one that breaks naive systems, and it is not rare.

Registration 21700900 covers three people: Werner Botha, number 512, running the 10 km; and Ruan and Ruben Kruger, numbers 1016 and 1017, both in the 5 km. One person entered all three on one evening with one card and one email address, and one of the three has no email address of his own on the entry at all.

A system that treats an entrant as the unit sends three letters to one inbox, two of which are addressed to people who never see it, and then expects three separate codes to arrive at the table in three separate hands.

The slimming step says, on the way past:

```
2 registration(s) cover more than one entrant (5 bags in all): one letter and one code each, which
hands over every bag on the registration
   21700900: 3 bags in LTR 10, LTR 5
   21700110: 2 bags in LTR 5
```

One letter goes to the address the registration was made with. It lists all three — race number, name and class — and its QR code holds the registration, not a member number.

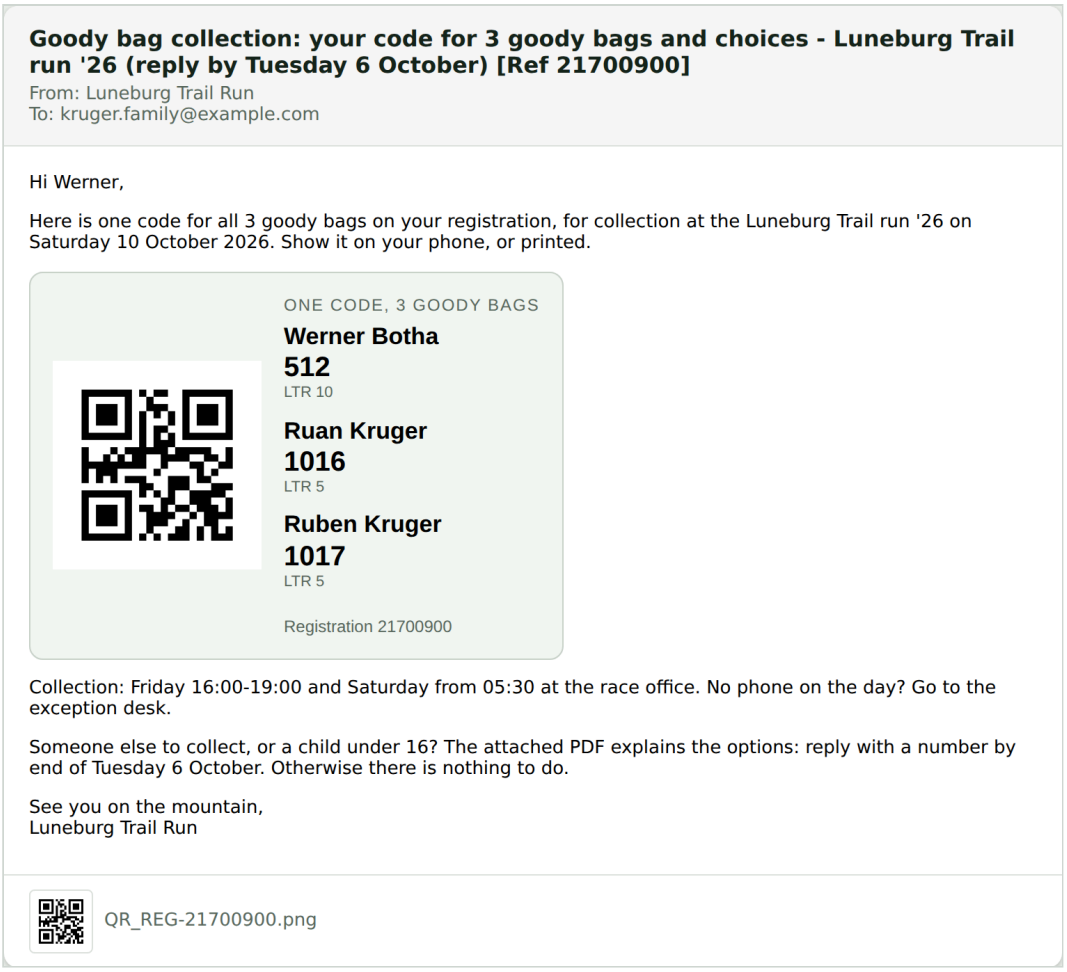


Figure 9. *One letter, one code, three goody bags across two classes. The three numbers are in the letter in figures as well, because the phone is not always the thing that survives the morning.*

At the table, that one scan puts the whole registration on the volunteer's phone:

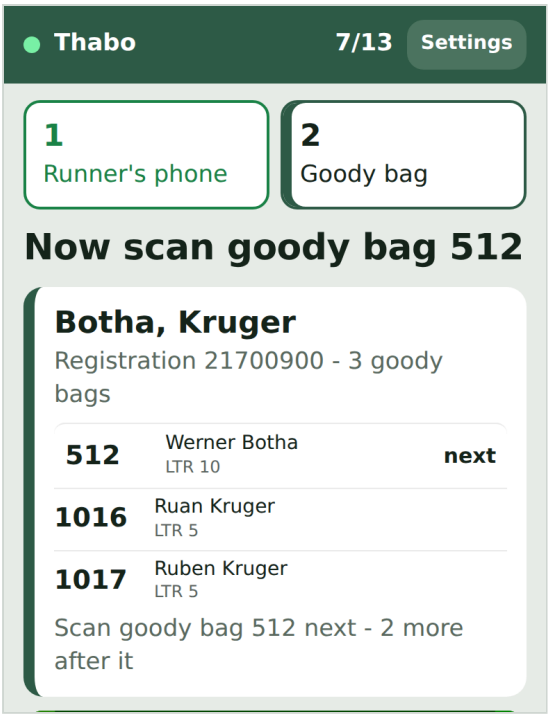


Figure 8. *The same registration as the phone sees it. All three bags, which one to fetch first, and how many are left after it.*

The volunteer then scans the three bags in whatever order they come to hand; after each one the list updates. Scan a bag belonging to somebody outside the registration and the phone refuses it by name, which is chapter 21.

And note the classes. Werner is in the 10 km, the Krugers in the 5 km. The bags are on different tables, at different ends of the aisle. The registration does not care, and neither does the phone.

None of this is compulsory, and the tick box that governs it starts in the off position. *Ignore the registration links* is the plain way and the default: everybody separate, one letter and one code and one bag each. Anke unticks it, which is what this chapter describes, and the argument for unticking it is arithmetic. Sixty registrations on the Luneburg export cover a hundred and fifty-one bags. Left plain, ninety-one people who currently arrive on somebody else's letter would each join the queue in their own right, and every one of them without an email address of their own would need their invitation printed and handed over at the desk. Thirty-eight of them have no address.

What the tick box really decides is not the letters. It is whether the phone knows a registration has three bags on it, and therefore whether anything in the building notices when somebody leaves with two. This year the question was settled for her by the next chapter: a rule about who collects for a group has to know who the group is, so the links are kept whatever the box says.

8. Sixteen and Over

The committee changed one more thing in September, and it started with a sachet.

Johan tells it at the meeting where it was decided. Last year a runner in the 10 km found a Tailwind sachet in his goody bag, ate the chocolate bar first, and then tipped the electrolyte powder straight into his mouth and chased it with a sip of water,

because it said *electrolytes* and he was thirsty. At the first water point he found out that the same powder, mixed properly, was in the urns. He finished. He does not recommend it.

"He is fifty-four," says Johan. "Now give that bag to a nine-year-old who opened her own email on her mother's tablet and collected it on her own."

Because that is how it had worked. A child with an email address on the entry had a letter of her own and a code of her own, and the code opened her bag whether or not an adult was anywhere near the table. Nobody had decided that. It had simply followed from treating every entrant the same way.

Then Thabo, who is the only person on the committee young enough to know, said the thing about email. Children's phones are put to sleep at night by their parents and wake at seven, and collection opens at half past five. A school account will quietly refuse mail from outside the school. And most of the teenagers he knows open their email when a teacher tells them to and not otherwise. A child's inbox, he said, is the least reliable way there is of getting a child their bag.

So the rule now fits on the same card as the other one:

> ***The person who registered, or anyone on the registration aged 16 or over, collects the group's bags. A child's own code opens nothing.***

The slimming step works out everybody's age on race day from the date of birth, and who registered each registration from the indemnity, and writes both into the working file, beside one more column with a single letter in it — *May Collect*, Y or N — and a sentence saying why not.

The rule is a default, not a sentence passed on anybody. In the last letter before the race the person who registered can change it once, by replying with a number:

1. A group member under 16 may collect for the group. Give their name.
2. Only I, the person who registered, may collect. All other codes on my registration will be turned away.
3. Someone who entered the race, but is not on my registration, will collect for my group. Give their name and race number.

The letter goes on the Sunday, and replies close at the end of the Tuesday. That leaves the Wednesday to read them and the Thursday to reprint whatever they change, which in a race week is not slack. It is the only margin there is.

The letter itself stays short, because most of the people reading it need to do nothing at all. Johan insisted on that after reading the first draft aloud to his neighbour, who stopped him at the third heading. The rule, the choices and the note about the sachet go into a one-page PDF attached to the letter of the person who registered; the email says one sentence about it.

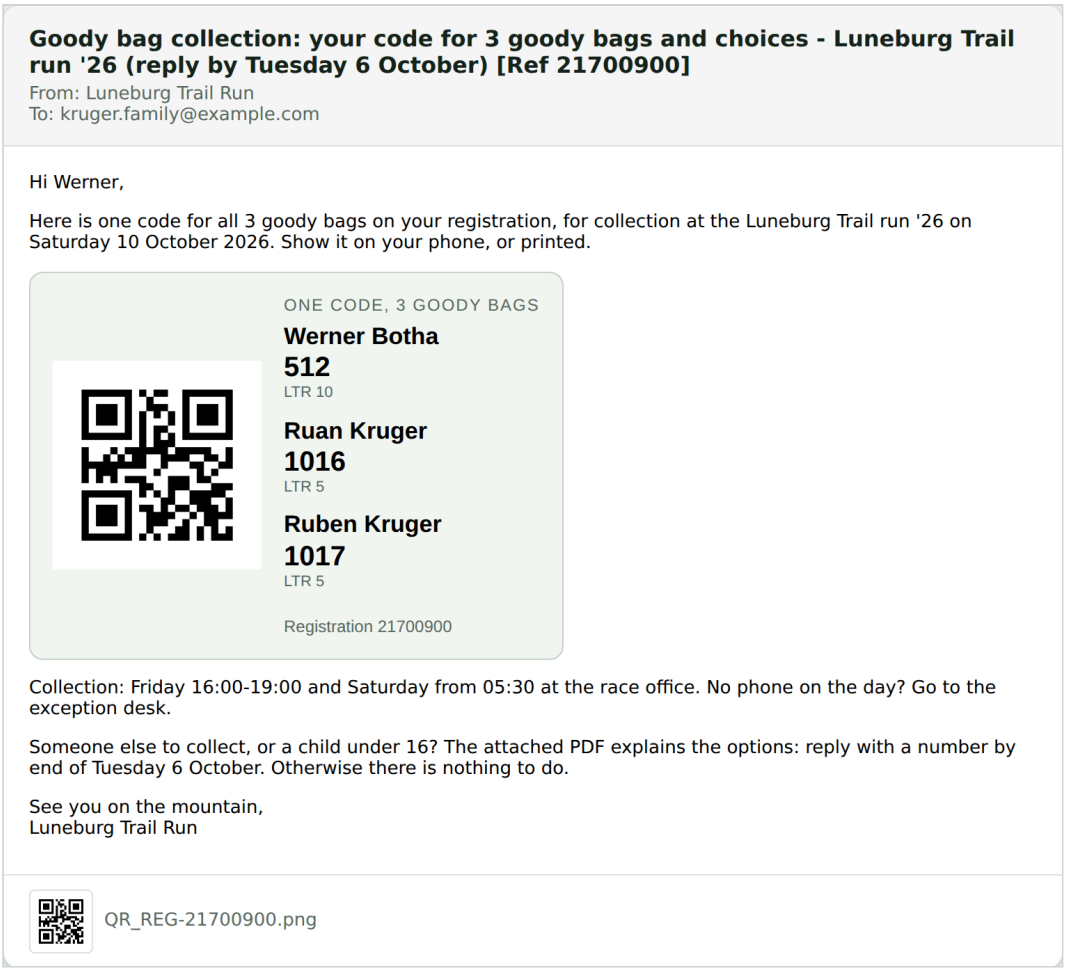
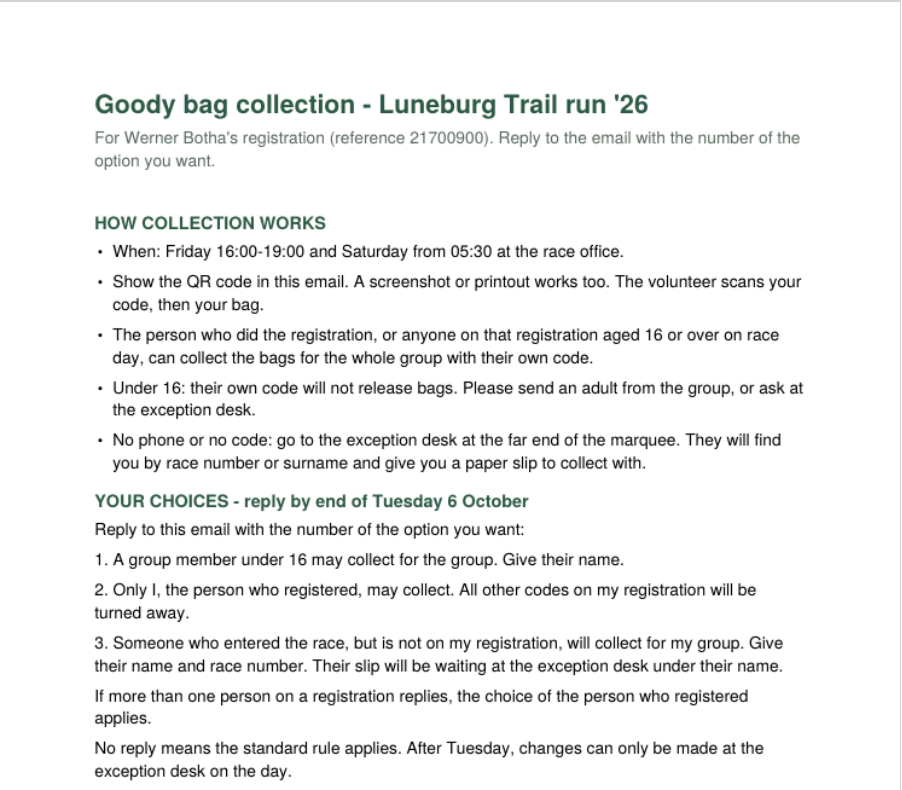


Figure 9. *The Kruger letter as it now goes out. One sentence, the code for all three bags with each name beside it, the race number underneath and the class in small print, when collection is, and one line pointing to the attachment for anybody who wants something different.*



Goody bag collection - Luneburg Trail run '26

For Werner Botha's registration (reference 21700900). Reply to the email with the number of the option you want.

HOW COLLECTION WORKS

- When: Friday 16:00-19:00 and Saturday from 05:30 at the race office.
- Show the QR code in this email. A screenshot or printout works too. The volunteer scans your code, then your bag.
- The person who did the registration, or anyone on that registration aged 16 or over on race day, can collect the bags for the whole group with their own code.
- Under 16: their own code will not release bags. Please send an adult from the group, or ask at the exception desk.
- No phone or no code: go to the exception desk at the far end of the marquee. They will find you by race number or surname and give you a paper slip to collect with.

YOUR CHOICES - reply by end of Tuesday 6 October

Reply to this email with the number of the option you want:

1. A group member under 16 may collect for the group. Give their name.
2. Only I, the person who registered, may collect. All other codes on my registration will be turned away.
3. Someone who entered the race, but is not on my registration, will collect for my group. Give their name and race number. Their slip will be waiting at the exception desk under their name.

If more than one person on a registration replies, the choice of the person who registered applies.

No reply means the standard rule applies. After Tuesday, changes can only be made at the exception desk on the day.

Figure 10. *The attached PDF: how collection works, the three choices with the deadline, and the sachet.*

Ruan Kruger, who is twelve, has an email address on the entry: his school's. He gets a letter too, and it is the most important letter in this chapter, because of what is not in it.



Goody bag collection: who collects your bag - Luneburg Trail run '26 [Ref 21700201]

From: Luneburg Trail Run
To: ruan.kruger@school.example.org

Hi Ruan,

Werner Botha collects your goody bag at the Luneburg Trail run '26 on Saturday 10 October 2026. There is nothing for you to bring.

RACE NUMBER
1016
Ruan Kruger
LTR 5

Collection: Friday 16:00-19:00 and Saturday from 05:30 at the race office. No phone on the day? Go to the exception desk.

The Tailwind sachet in the bag is meant to be mixed with 500 ml of water. Please don't take it dry: it can upset your stomach. Mixed electrolyte drink is also waiting at every water point.

See you on the mountain,
Luneburg Trail Run

no attachment

Figure 11. *Ruan's letter. One sentence saying who collects his bag, his number, his name and his class, and the note about the sachet. No code. If his school's mail server swallows it, nothing is lost: there was nothing in it to lose.*

"No code at all?" says Johan.

"Nothing to scan, nothing to forward, nothing to screenshot," says Anke. "If he turns up on his own, the desk sees a twelve-year-old and his father's name. That's a conversation, not a handover."

9. One With No Number, One With No Address

The late entry is imported anyway, because Anke ticks the box that says so, and alert 14 names him every time anything runs. He gets no code and no race number in his letter. What he gets is a letter that says his application needs custom handling, and asks him to see the organisers on arrival.

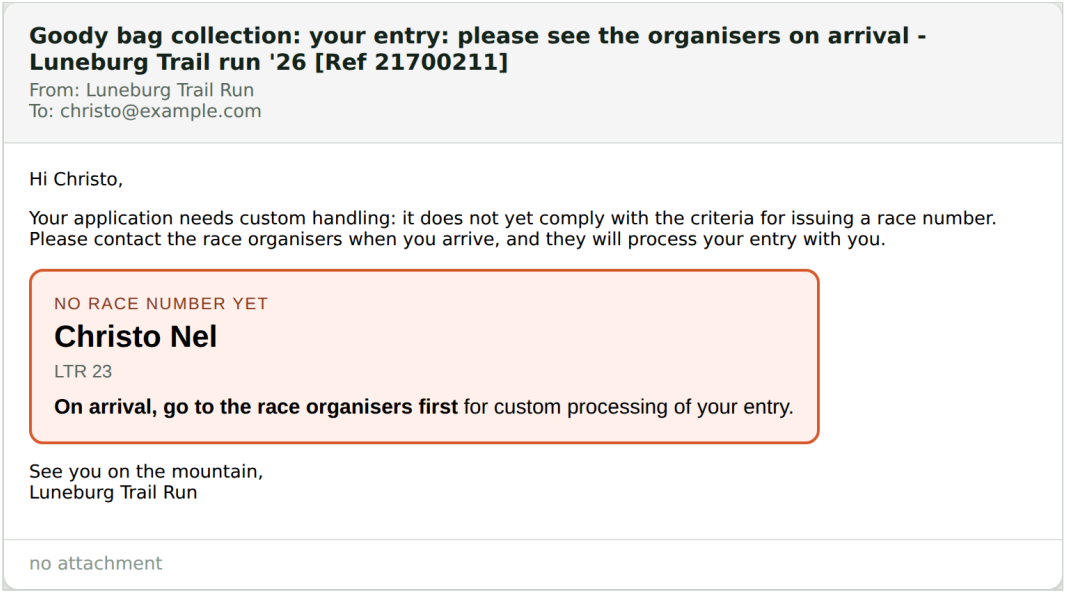


Figure 12. *No number, no code, no goody bag waiting — and an orange panel that says where to go instead. The one thing this letter must not do is look like everybody else's.*

He has no square on the table and nothing he can collect, and he cannot reach the start list until somebody at the late entry desk has dealt with him.

The other exception in that same run is quieter, and worth more than it looks:

```
##### ALERT 18 #####
1 registration(s) have no email address
  21700118: Mevrou Steyn
Nothing is sent for them. Print their invitations instead, or add an address in EntryNinja and export
again.
Action: Print their invitation instead and hand it over at the desk, or add an address in EntryNinja,
export again and slim again.
See the setup guide, Alerts, number 18.
##### END ALERT 18 #####
```

Mevrou Steyn is eighty-one, runs the 10 km every year at a pace that embarrasses people half her age, and has never had an email address and has no intention of getting one. The mailer does not invent one, does not send to a blank, and does not skip her silently. It stops, names her, and tells you to print her page instead.

Printing her page is not a consolation prize. It is the same code, on paper, and it goes through the desk exactly like everybody else's.

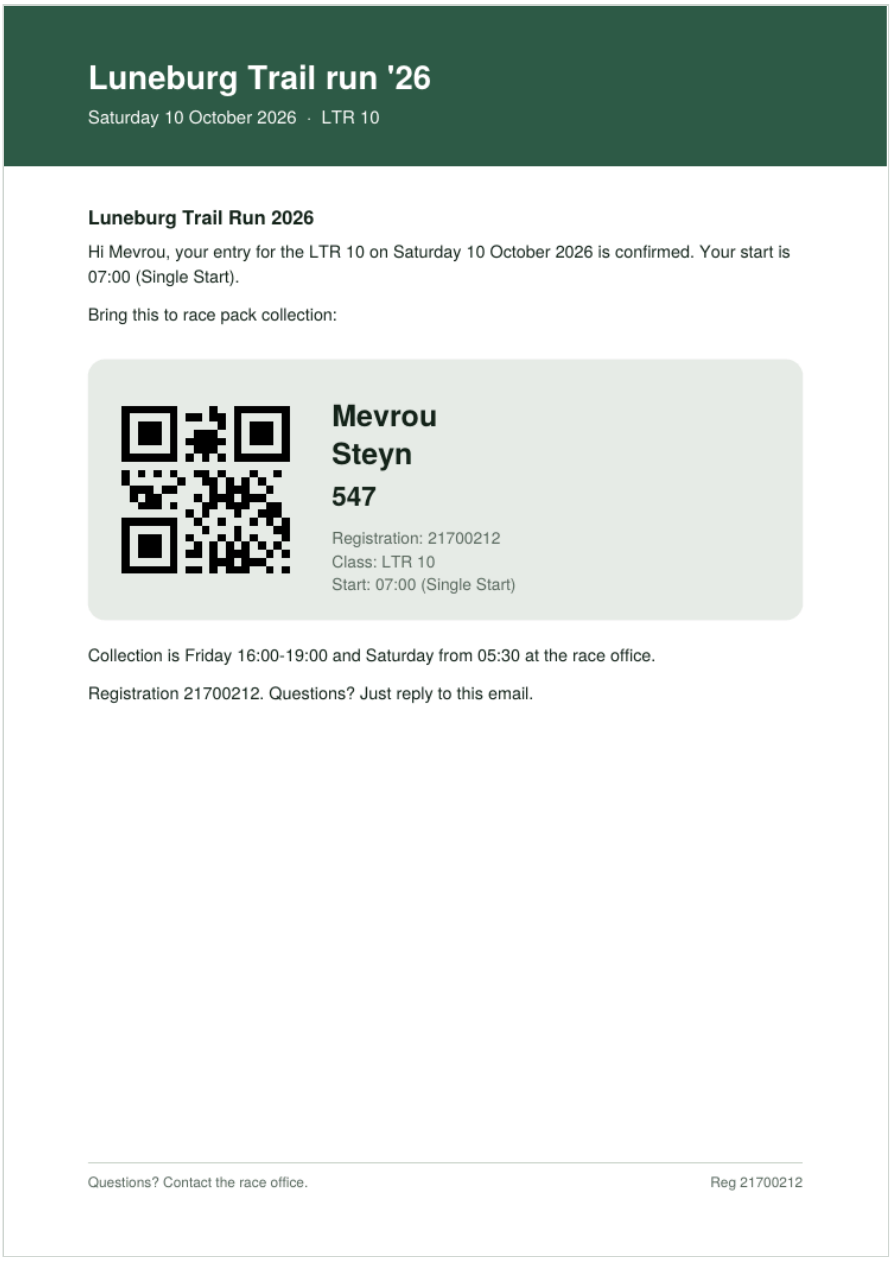


Figure 13. *Mevrou Steyn's invitation, printed. Her code is on it at a size a thumb can hold up in a queue, and the desk scans the paper without knowing or caring that it is paper.*

10. Three Letters, Not Three Hundred

The letters go out on the Sunday, and on the Saturday before, Anke sends three of them to herself.

Not three hundred. Three. The screen will not take a fourth: *4 given: a test send takes at most three race numbers or surnames*. Send yourself twenty and you read the first one. Send yourself three and you read all three.

She picks them so that each is a different kind: one ordinary runner, one registration with three bags on it, and one surname rather than a number, because a late entrant has no number to type and that is the whole reason the box takes surnames.

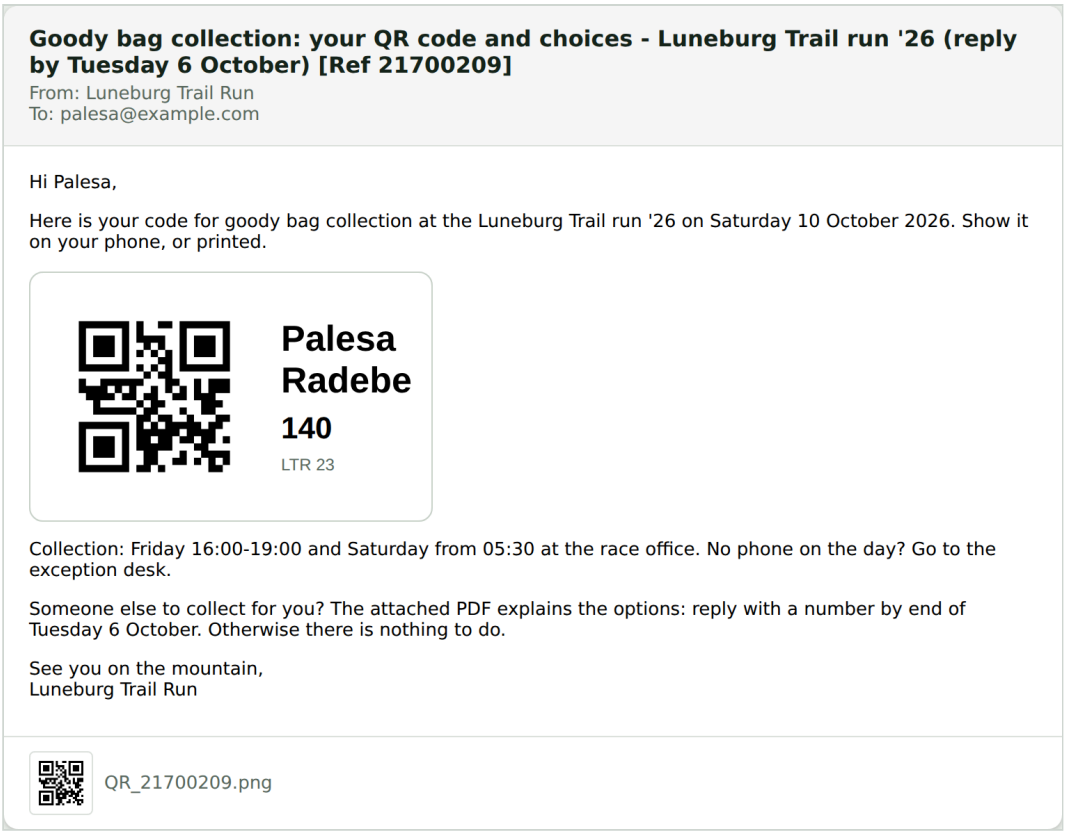


Figure 14. *The ordinary one, for a runner who entered alone. One sentence, the code with her name beside it, when collection is, and one line about the attachment. Her PDF offers only option 3: on a registration of one there is nobody under sixteen to name and nobody else to turn away.*

None of the three touches the sent log, so the real send on Sunday is unaffected.

Anke reads the first one standing in the kitchen with one thumb, which is the test that matters, because that is how it will be read on a Saturday morning in a car park with a child asking for the toilet.

11. Six Replies

By Wednesday morning the race mailbox has six messages in it that were not there on Sunday, and Anke does not open any of them.

She presses *Read reply emails now* on the race desk instead, and the reply reader goes through the unread mail on its own. She runs it again on Thursday morning, for anything that came in overnight; the second run is the one that counts:

```
APPLY kruger.family@example.com       registration 21700900 (Werner Botha): 1 - Ruben Kruger (under
16) may collect for the group
APPLY sibusiso@example.com            registration 21700105 (Sibusiso Ngcobo): 0 - standard rule
LEAVE lerato@example.com              option 3, but no race number or name of another entrant found
LEAVE palesa@example.com              option 3, but no race number or name of another entrant found
LEAVE marius@example.com              arrived Thu 08 Oct 08:15, after the reply deadline
2 reply(ies) read clearly, 3 left in the inbox, 1 other message(s) not touched
```

It is a deliberately timid program. It applies a reply only when there is nothing left to decide: the reference in the subject line finds the registration, the sender is an address on it, the reply came before Tuesday midnight, and there is exactly one option in it, on a line of its own or after the word *option*. Werner's *Option 1 please. Ruben can collect for us* names the one child it could mean, and it is applied. Sibusiso's *the standard rule*

is fine is applied as no change at all. The braai club's question about parking is not a reply to the letter, so the reader does not touch it — not even to mark it read.

The other three it leaves exactly where they were, unread, and says why:

```
##### ALERT 22 #####
3 reply(ies) left in the inbox, unread
  lerato@example.com: option 3, but no race number or name of another entrant found
  palesa@example.com: option 3, but no race number or name of another entrant found
  marius@example.com: arrived Thu 08 Oct 08:15, after the reply deadline
They are also on the Replies sheet of collection_choices.xlsx, with the reason.
Action: Open each one in the mailbox and enter the choice by hand on the Collection choices screen (it
says what the reader could not decide). Late replies are for the exception desk on the day.
See the setup guide, Alerts, number 22.
##### END ALERT 22 #####
```

Lerato's *Maybe option 3? My sister might fetch it, I'll confirm* is a question, not a choice. Marius's came on Thursday morning, two days after the deadline, and late replies are for the exception desk on the day. And Palesa's *My friend Gerda will collect for me, she runs the 10* is the kind a person can settle in a minute and a program should not try: a first name and a distance is not a race number.

So Anke phones Palesa. The *Collection choices* screen lists the replies the reader left behind. She finds Palesa's registration, picks option 3, types Gerda van Wyk's race number — the screen puts Gerda's name beside it, so a wrong digit shows — writes down why, and saves.

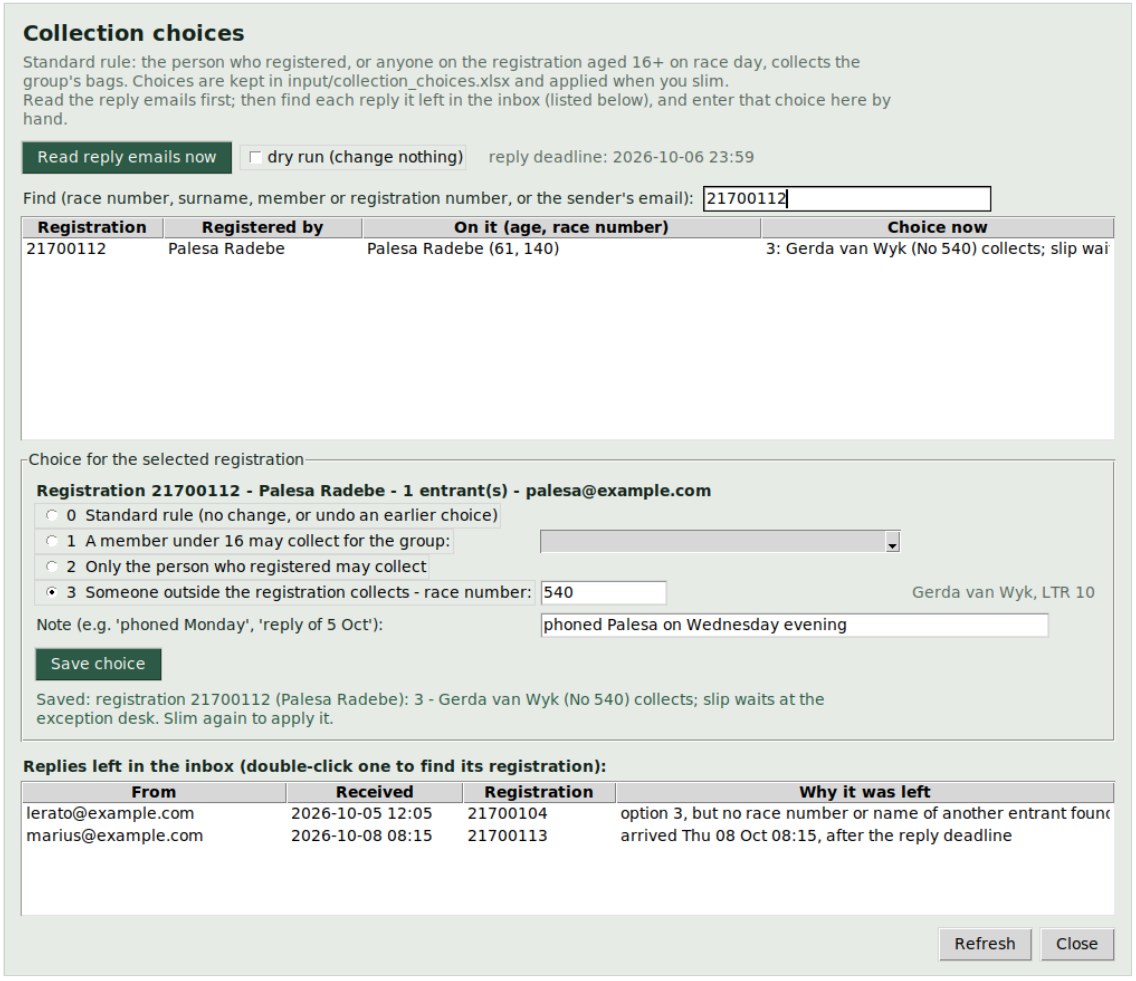


Figure 15. *The Collection choices screen just after saving. The reader's leftovers are listed at the bottom with the reason each was left. A choice entered here counts as the registrant's own, so a later reply from anybody else on the registration cannot overturn it.*

Lerato never confirms, and her registration stays on the standard rule, which is what silence means. Marius is told at the desk. Then Anke slims the export again, because slimming is what turns the choices into who may collect.

And slimming now prints a sheet nobody has to ask for. Every time the export is slimmed with the rule on, and every time *Import and start* is pressed, the exception desk gets four pages of its own:

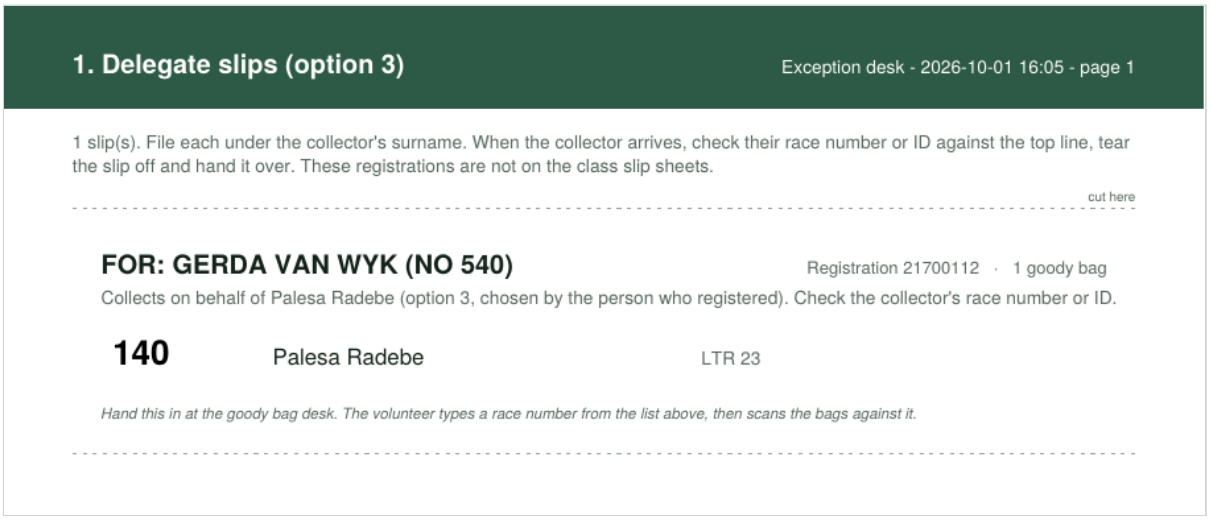


Figure 16. *The first page: a slip already printed for Gerda van Wyk, her name and number across the top and Palesa's bag underneath. Palesa's registration has come off the 23 km slip sheet, so there is still exactly one slip in the building that releases her bag.*

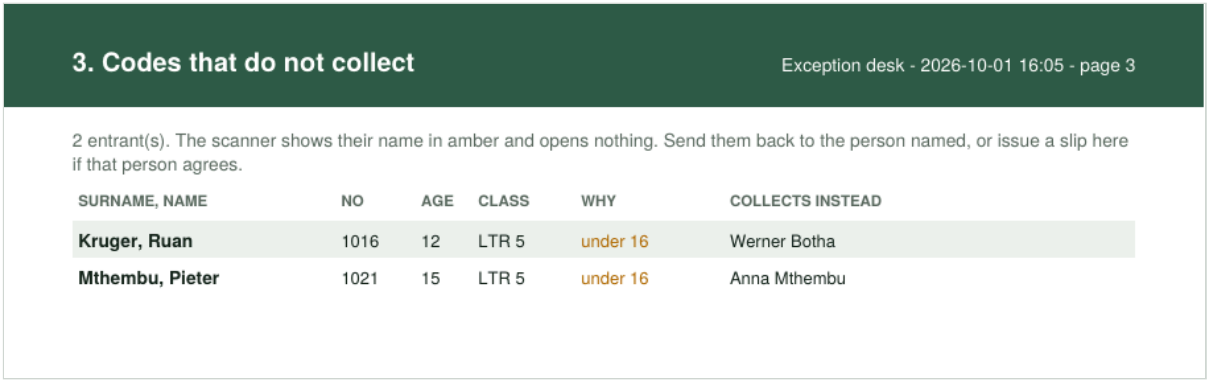


Figure 17. *Further in: every entrant whose own code the scanner will refuse, and who collects for them instead. Pieter Mthembu is fifteen and Ruan Kruger is twelve. Ruben, who is thirteen, is not on the list, because his father named him.*

12. The Letters Nobody Sends

The 65 km letters are not sent by the system at all, and the menu knows it.

Twenty-seven runners, and no two of them are doing the same thing. Some are camping Friday and Saturday; one has a hostel bed; one is bringing a support crew of four; two have never run further than a marathon and need the cut-offs explained gently and at length. The kit list alone runs to half a page.

So that class is set to external, and instead of letters the system writes twenty-seven pictures — one card per registration, named by race number, with a spreadsheet of names and file paths beside it — and Anke writes the letters herself in Word and merges the cards in.

The card holds the runner's code, not the bag's. It did not, once. Thabo found that on a dry run in September: he scanned a card and the phone said, quite correctly, that is a goody bag label, not the runner's code. Now the system works it out from how the class is handed over, and every class is handed over the same way, so every card carries the runner's code.

There is one more thing the menu does here that looks like an absence. The button marked *Invites (PDF)* is not on the screen at all until a class has a Word letter attached to it. No letter, no invitations — because an invitation booklet with nothing in it but a merge field is a stack of paper nobody wants. When Anke attaches the 65 km letter, the button appears. When she takes it off again to test, the button goes.

13. The File That Was Open

On the Thursday evening, a week out, Anke has the 65 km RaceClocker file open in Excel to see who is on it, and runs the export again without closing it.

What used to happen was a wall of Python ending in *PermissionError*, which means nothing to anybody at half past five on a race morning. What happens now is:

```
##### ALERT 15 #####
Could not write or read a file: it is open in another program
  output/raceclocker_LTR_65.xlsx
Windows said: Permission denied
Action: Close the file named above in Excel or wherever it is open, then run the
same step again. Files written earlier in the run are complete; running again
rewrites them all.
##### END ALERT 15 #####
```

She closes Excel and presses the button again, and all four files write in eight seconds.

"It's Windows holding the file," she tells Thabo later. "Not us. But the person reading the screen at half past five in the morning doesn't care whose fault it is. They want the sentence that ends with *close it in Excel*."

14. The Practice Round That Would Not Repeat

Thabo Mdlalose is nineteen, studying IT in Pietermaritzburg, and home for the weekend to learn the phone. On the Tuesday before the race they run the proof properly. The proof page is the dry run made visible: for each registration the runner's code, then the goody bag label at its real size, alternating left and right down the screen so the phone has to travel each time, exactly as it will along the table.



Figure 18. *Step one, on the left: the registration's code, with all three numbers it covers and the name beside each. The page says what the QR actually holds, so a wrong code is caught here and not at the table.*



Figure 19. *Press space and step two appears on the right, at the true size of the 102 by 47.2 mm label. Three bags on this registration, so this is one of three, and the page says which.*

The second time round Thabo goes too fast. He confirms Luca Beneke's code, presses space twice, and holds the phone over the next card, which is his brother Oliver's code. Last September that is where it would have gone wrong: the phone was still holding Luca, the new code meant nothing to it at step two, and the next label Thabo scanned was Oliver's — *Wrong bag for this runner: Luca Beneke. This bag is printed for Oliver Beneke* — two names on a screen where the page in front of him showed only one. Now the phone says what it is holding before anything is sent.



Figure 20. *A runner's code at step two. The phone is still on the last runner, whose bag was never scanned, and it says so on tab 1: tap it to start with the new runner. The bag tab stays empty, and nothing has been sent.*

"The bag for the one before was never scanned," says Anke. "That's what it's telling you. On the day that means a runner walked off without their bag, or you skipped one. Either way you want to know now, not when the next label comes up with the wrong name."

They find one surname with a missing letter and nothing else. For two hundred and sixty-two runners they would otherwise have checked a handful and hoped.

The proof page tests the codes. It does not test the table: the walk from the runner to the bag, the number turned into a place, the phone held over one label among thirty. So on the Wednesday, after the first thirty-two runners have been scanned off the screen, the living-room floor becomes the marquee. *Import and start* has already printed the paper to carry on with: the next sixty runners, thirty-three to ninety-two, in the proof page's own order, twelve to a page. Five pages headed MAIL, which Anke holds up the way a runner holds out an email, and five headed GOODY BAG, which Thabo spreads along the carpet like bags along a table.



Figure 21. *MAIL: the runners' codes as the emails carry them, surname, first name and race number beside each.*



Figure 22. *GOODY BAG: the same runners in the same places, so card five on the third MAIL page belongs with card five on the third GOODY BAG page. (Both from a full-size generated list of 259 entrants numbered like Luneburg's; the test list of thirteen has no runner thirty-three, and says so.)*

"Why not just keep going on the screen?" says Anke.

"Because the screen never makes you walk," says Thabo. He scans sixty runners and sixty bags in twenty-two minutes, and the wall on the laptop fills in with colour, one square at a time.

Then Thabo says, "Let's do it again, faster. I want to see how long it takes."

He scans the first card and the phone says *Already collected*, with Sibusiso Ngcobo's name and the time, four minutes earlier.

"It remembered."

"Of course it remembered." Anke is laughing. "It's the real desk. You just checked people in. If we'd left it like that, on Saturday morning the phone would have told you every one of them had already collected, and you'd have thought the laptop was broken."

Which is exactly what happened to her in September, and cost her an hour hunting for a fault that was not there. So there is a way out of it now: stop the server, clear the check-in record, and the working file goes back to the way the slimming step left it, and the record the server keeps is forgotten.

```
Wrote input/race_day.xlsx
  12 not collected, ready to scan again
  1 with no race number: left not collected
  forgot the check-in record: output/checkin_files (4 file(s)) deleted
```

"Practice rounds are real rounds," she says. "That's the whole sentence. Practise as often as you like, as long as you clear it afterwards. And on Saturday you never clear

anything."

Thabo clears it, starts the server again, and watches it say *Loading entries* rather than *Resuming from*, which is the difference between a fresh desk and yesterday's.

He also, being nineteen, deletes the check-in folder while the server is running, to see what happens, and reports with some disappointment that the program simply makes the folder again, writes the next file into it and carries on:

```
output/checkin_files was missing: created again
CHECK-IN 21700203 Sibusiso Ngcobo bag=BAG-21700203 -> checkins_00002_20261001-160549.xlsx
```

Anke, who lost twenty minutes of her life to exactly that at ten to eleven at night in September, says nothing at all and smiles at the kettle.

15. The Thursday Drill

On the Thursday Anke makes him do the one thing nobody wants to do on the day.

"Pretend the power's gone. Marquee's dark, laptop's flat, phones can't see anything."

Thabo opens the exception desk on the registration PC and presses *Scanning has stopped*, typing the reason into the box beside it.

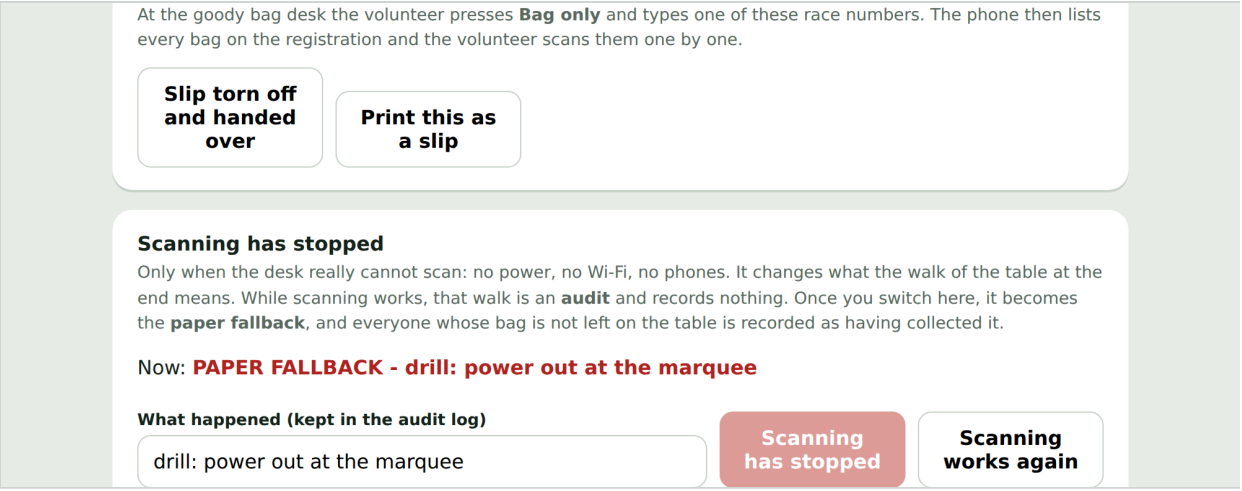


Figure 23. *The one switch in the whole system that changes what a later action means. It takes a reason, the reason goes in the audit log, and only the PC at the desk can press it.*

The leftover app on his phone, which had been green and said *checking the table against the record*, turns red within ten seconds. A button appears at the bottom that was not there before.

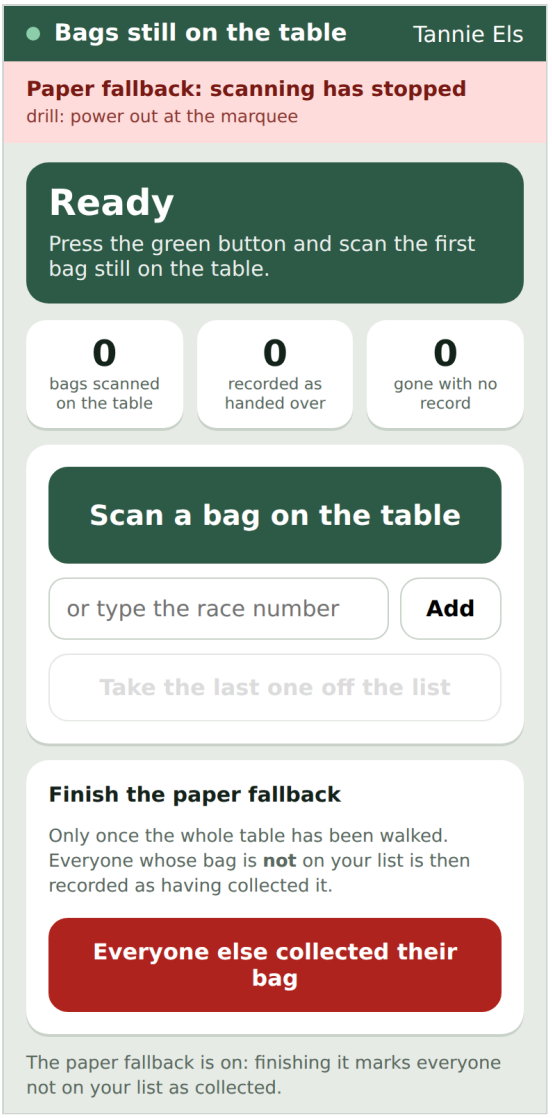


Figure 24. *Same app, same walk, same green scan button. What has changed is the banner, and the red button at the bottom that only exists while the banner is red.*

"That's the difference," says Anke. "Same walk of the table, same green button, same scanning. What changes is what it means. Green, it's an audit and it records nothing. Red, the table is the record."

Then Thabo declares scanning working again, and the red goes away.

"And if it happens for real," he says, "who decides?"

"Whoever's standing at the PC. Not you, holding a phone in a queue. That's why the button is over here and not over there."

Part Three: The Paper

16. Five Sheets

The slips print on the Wednesday, five files, and the partition is the point of them.

```
Wrote 5 file(s) of tear-off slips, 262 goody bag(s) in all:
  output/slips_LTR_5.pdf             17 slip(s),   19 bag(s), 3 page(s)
  output/slips_LTR_10.pdf            81 slip(s),   85 bag(s), 13 page(s)
  output/slips_LTR_23.pdf           109 slip(s),  114 bag(s), 17 page(s)
  output/slips_LTR_65.pdf            24 slip(s),   25 bag(s), 4 page(s)
  output/slips_across_classes.pdf     7 slip(s),   19 bag(s), 3 page(s)

Every registration is on exactly one sheet: a slip torn off is gone for good,
which is what makes the paper the authorisation.
```

Four sheets for the four classes, holding the registrations whose race numbers all sit inside one class — including a family with three numbers in the 5 km, which is one slip with three lines on it, not three slips.

And a fifth sheet for the seven registrations like the Krugers, whose numbers are spread across classes and therefore belong to none of the four.

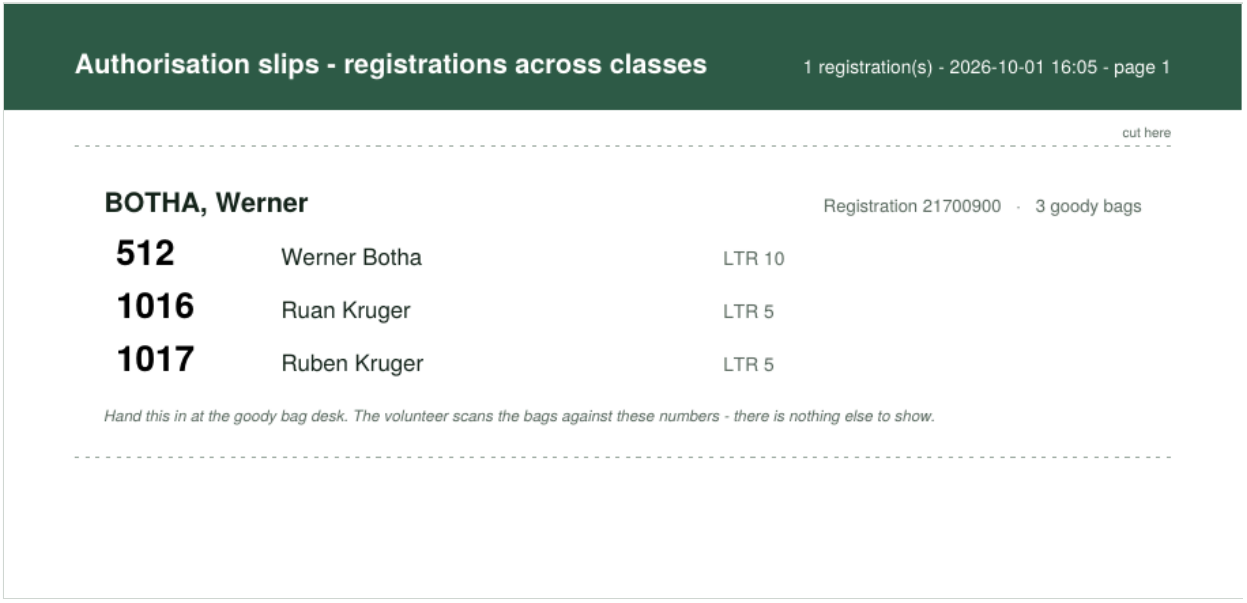


Figure 25. *The fifth sheet. One slip, three numbers, two classes — and a note saying which sheet it is on, because a slip that could be reprinted from two places would authorise twice.*

Every registration is on exactly one sheet. That is what makes a torn slip final: it is gone from the only place it ever was, and the gap shows. The one registration whose family chose somebody else to collect is on none of the five. Its slip is on the exception desk's own sheet, with the collector's name printed on it.

Each slip is a surname in bold, then one double-spaced line per bag — the race number in figures you can read across a marquee, the person's name, the class — and along the bottom, small: *Hand this in at the goody bag desk.* The volunteer scans the bags against these numbers — there is nothing else to show.

Tannie Marietjie, who has run the registration desk for eleven years and regards it as hers, holds a sheet up to the light and tears one off, and is satisfied.

17. The Desk That Says No

The exception desk is a trestle table at the far end of the marquee with the pastor behind it, five sheets of slips, a pair of scissors nobody uses because they tear better by hand, and a laptop showing one page.

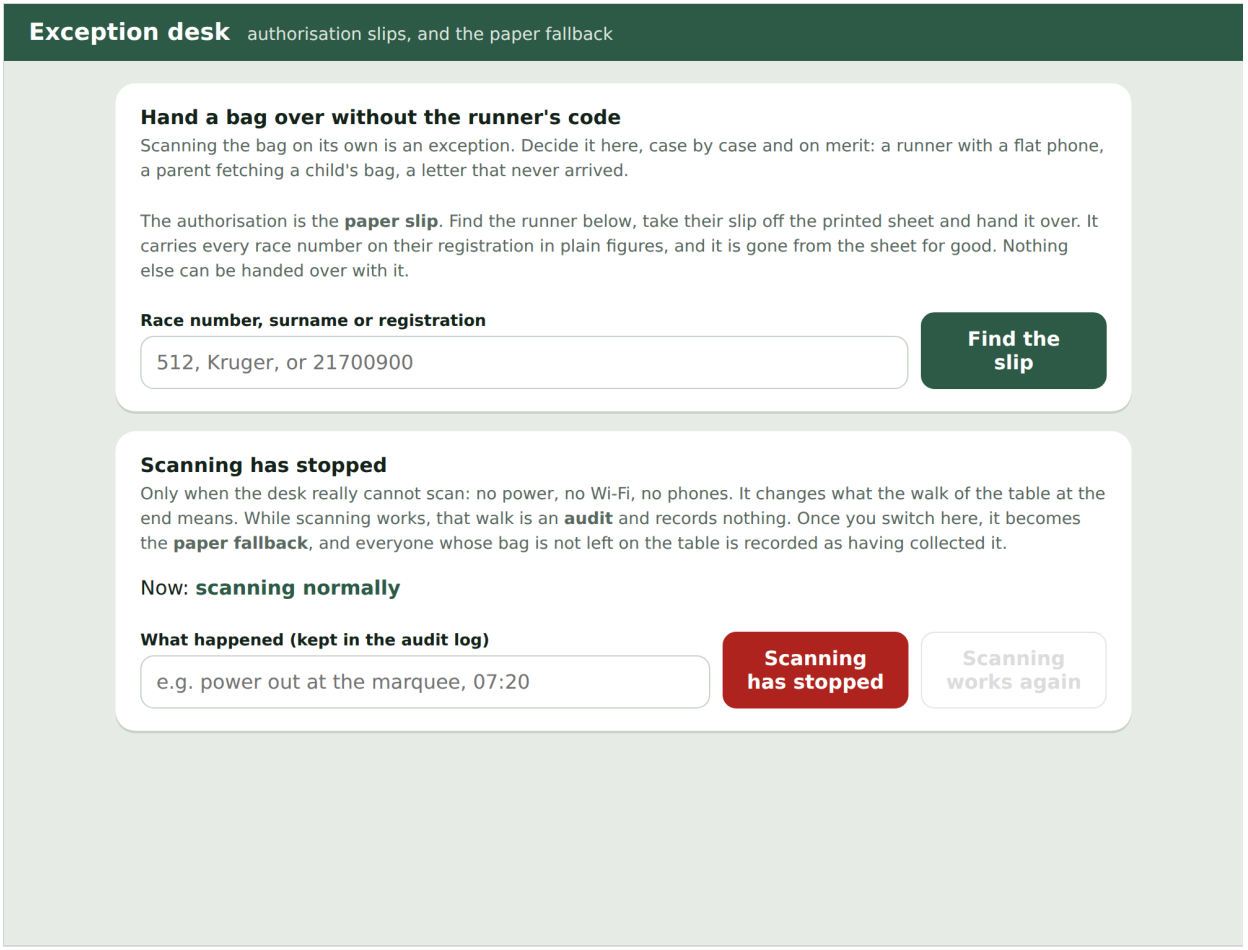


Figure 26. *The whole exception desk, in one screen. It answers exactly one question, and the question is written across the top of it.*

The desk does not need a code, which is the point, because the person standing in front of it has not got one. It takes a race number, or a surname, or a registration number, and it finds the registration behind them.

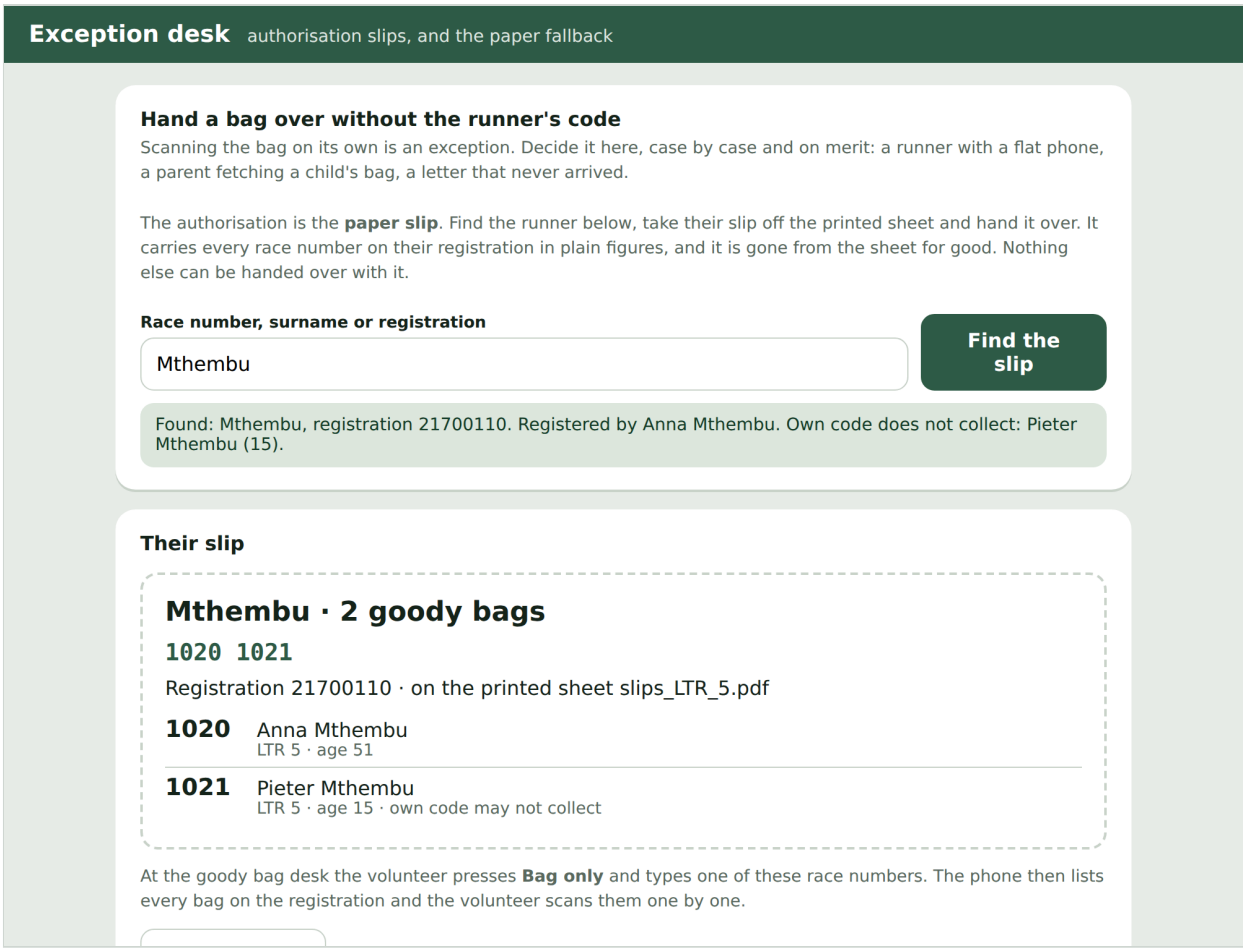


Figure 27. *A surname was enough. Two goody bags on one registration, both numbers in plain figures, which of the five printed sheets this slip must be torn from — and who registered, and that Pieter, at fifteen, cannot collect on his own code.*

The number and the name, which used to be the whole short-race process, are now this: a way of finding a human being in a list, at a table where a human being is standing. What comes out of it is not a bag. It is a strip of paper.

When the pastor tears it off, he presses one more button, and the tear goes into the audit log with a time on it.

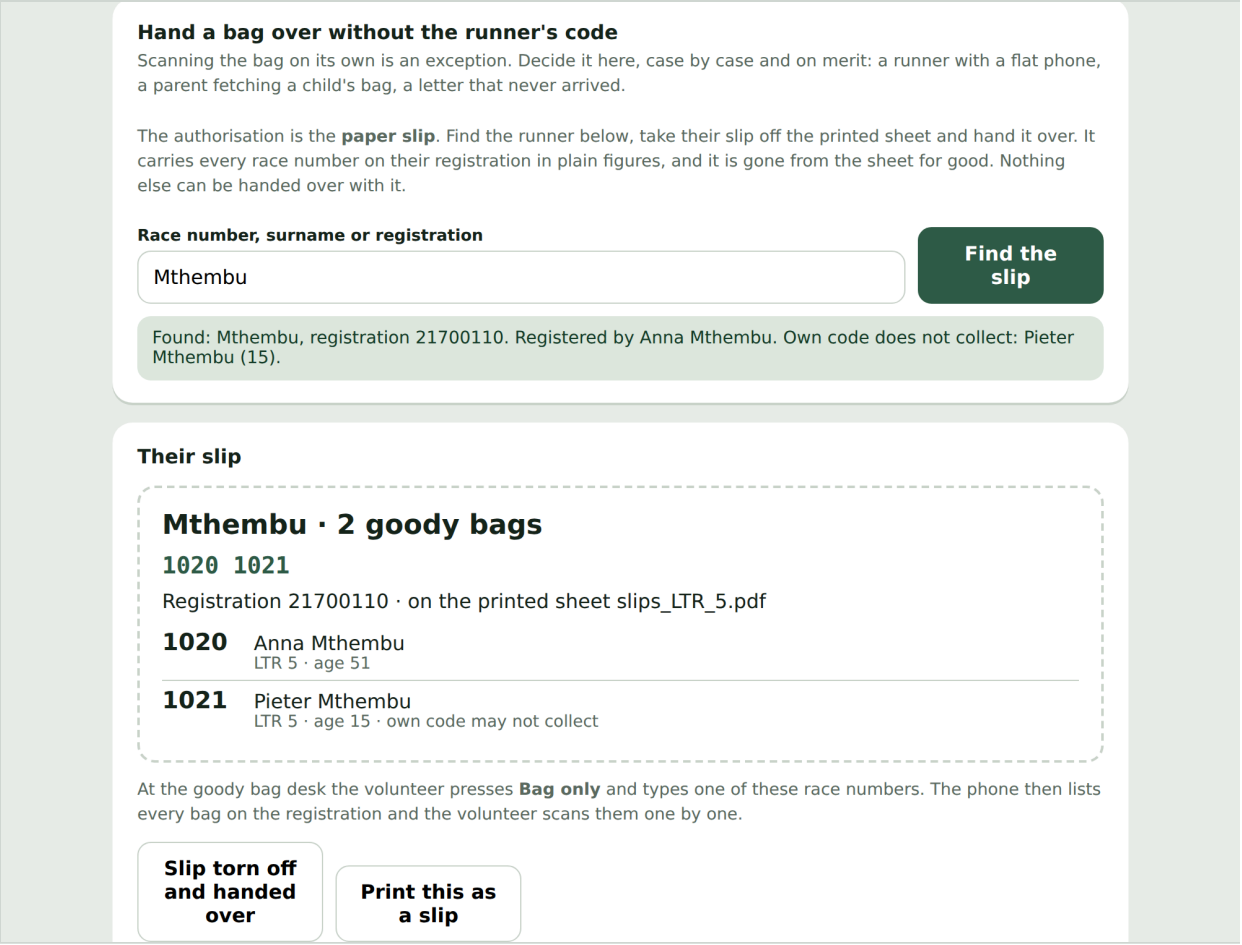


Figure 28. *Logged. The system can now tell an authorised bag-only handover from an unexplained one, which is the difference between an exception and a hole.*

"Why does that matter?" Thabo asks. "The paper's the lock either way."

"Because on Sunday morning somebody will count the gaps in the sheets and count the bypassed handovers in the record," says Anke, "and if those two numbers disagree, I want to know which way."

The desk also knows about the Krugers.

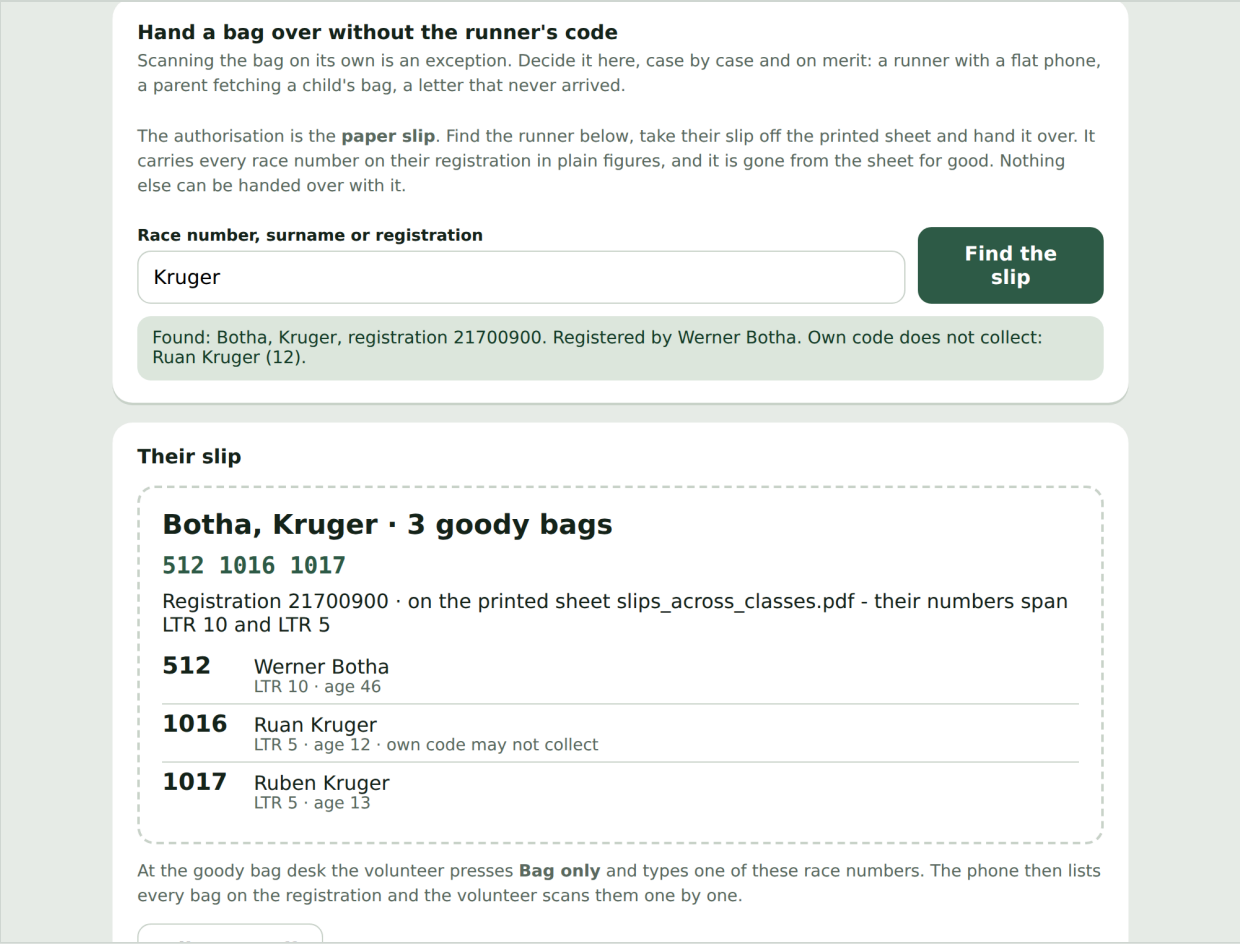


Figure 29. *A registration whose numbers span two classes is not on either class sheet. The desk says so, in the same sentence as the names, so nobody hunts through the wrong pile. It also says that Ruan's own code will not collect, and that Ruben's will.*

Then Thabo, being nineteen, tries to break the other end of it. What he finds is chapters 20 and 21.

Part Four: The Weekend

18. Three Buttons

Friday morning, and the whole desk comes up in one press.

The easy desk used to have exactly two buttons on it, because on the morning of a race nobody should be reading a menu. It now has three, and a small row underneath, and Anke defends every one of them. *Import and start* takes the export from input, cuts it down, makes the labels, the picture cards, the invitations, the collection list, the block map, the slips, the exception desk sheet and the proof page, starts the check-in server, and opens two browser windows. *Export all* is for the night. *Restart server* is for the morning after the night, which is chapter 27. And the row underneath, with a class picker and a button marked *Export this class*, is for the one class that has to leave before the others are finished.

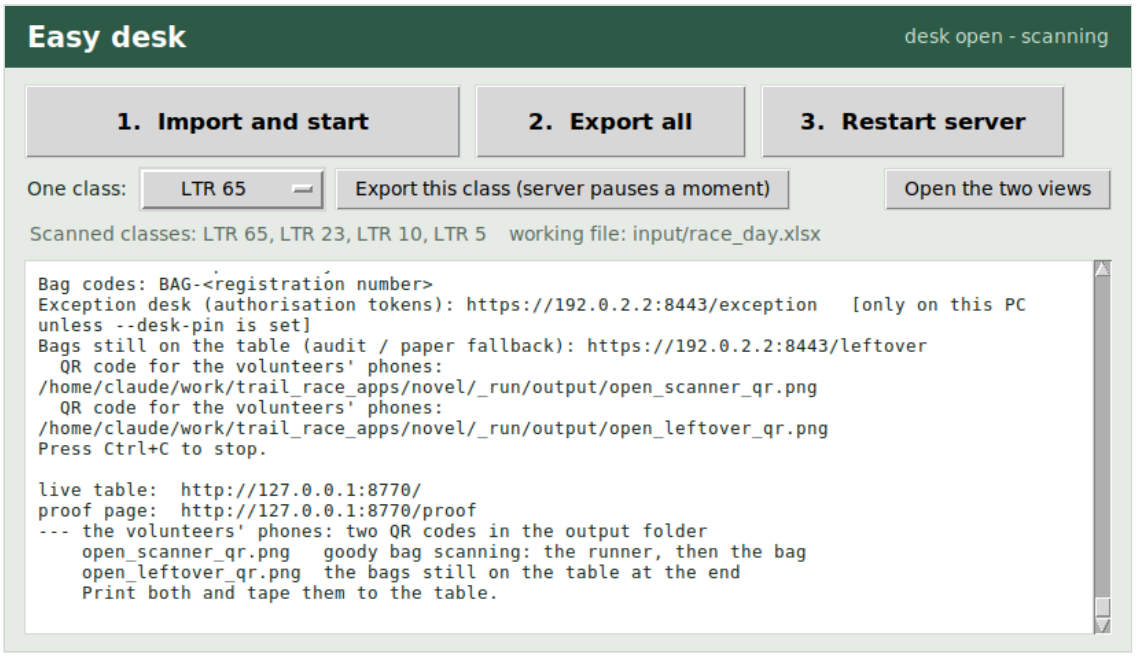


Figure 30. *The easy desk, just after Import and start. Ten lines of log, which is as much as anybody reads on a morning like this, turning red for anything that needs a person.*

The first of the two windows is the live table: the whole marquee from above, drawn the way it is laid out. One box per class and hundred, in race number order along the aisle; the front table above the aisle, units 0 to 4, and the back table below it, 5 to 9; and the aisle itself, a pale strip down the middle, shared out between the two volunteers with a dashed line where one hands over to the next.

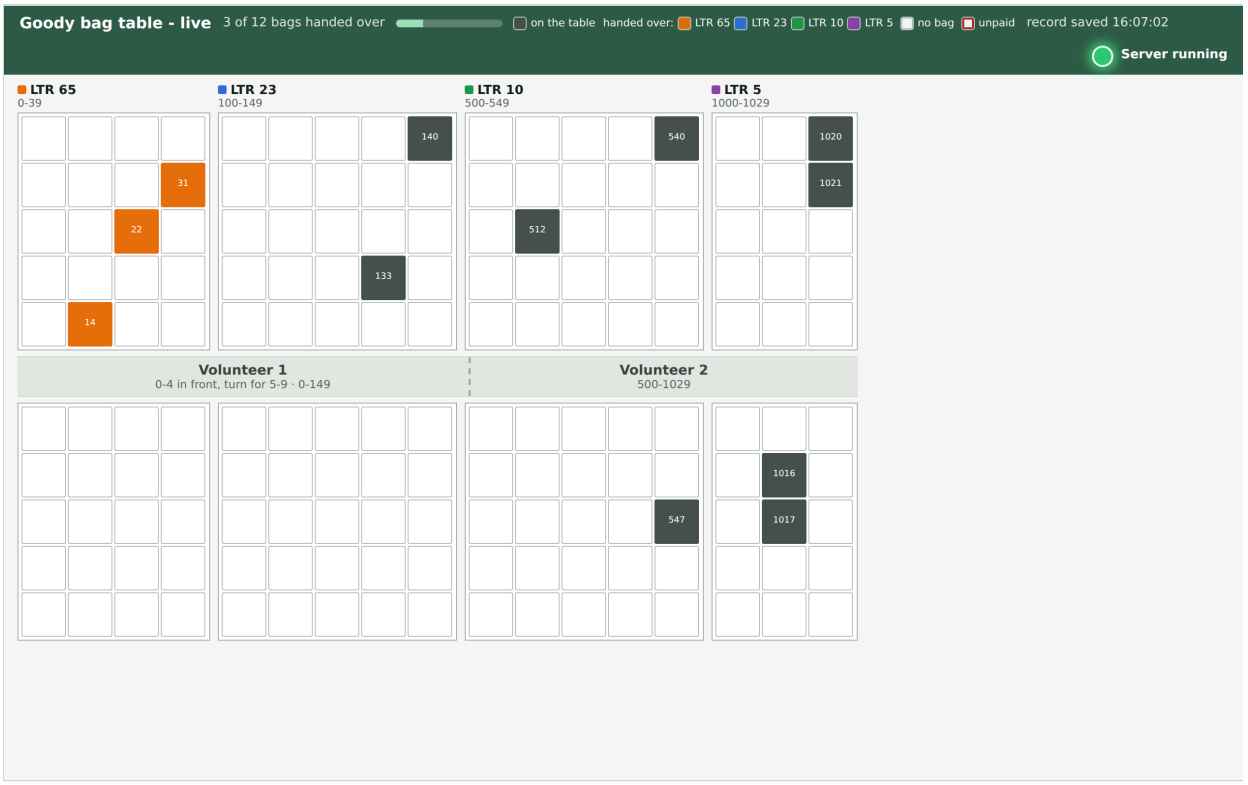


Figure 31. *The live table on the wall behind the desk, part-way through Friday. Dark grey is still on the table. A bag that has gone turns the colour of its own race: orange for the 65 km, blue for the 23 km, green for the 10 km, purple for the 5 km. An empty outline is a number nobody holds, and the round light in the corner says the server is running.*

Last year it was a list of squares in rows of ten, one class under another, and it told you who had collected but not where to look. This year it is the shape of the tables, so that a volunteer can look up from the aisle, find the grey square on the wall, and put their hand on the bag.

"Grey is still here," says Anke. "When a bag goes, it turns the colour of its race."

Oom Dawie looks at it for a while. "So at the end of the evening, if the 65 km box is all orange, everyone's got their bag."

"And the grey ones are the ones who haven't come."

"And that light?"

"That's the laptop. Green, the phones are talking to it. Red, they're not, and somebody should find out why before the queue does."

It refreshes itself every second and a half from the check-in record, so nobody has to press anything. By eight that evening the 65 km box is mostly orange, and Johan — who does not trust computers but trusts a picture — stands in front of it twice with his hands behind his back, counting grey squares.

19. Friday, Six O'Clock

The marquee goes up in a wind that tries to take it to Mozambique. By four the bags are out by the block maps, Thabo reading numbers and Tannie Marietjie placing them, and twice she reaches for a gap to close and twice he says "Tannie, the gap," and she withdraws her hand with the dignity of a woman who meant to do that.

The phone gets set up once, at the start of the shift, and then not touched again. It used to ask for a PIN as well as a name, and the PIN was written on a card on the

laptop, and the card was read out across the marquee to every new volunteer, which made it a password in the way that a front door left open is a lock. Now there is one field.

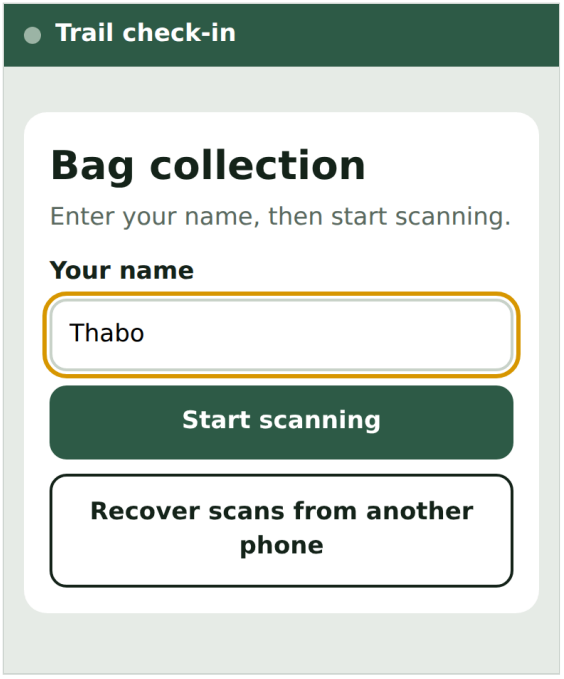


Figure 32. *One field and a button. The name is not decoration: every handover carries whoever was holding the phone, and on Sunday that is how a question gets answered.*

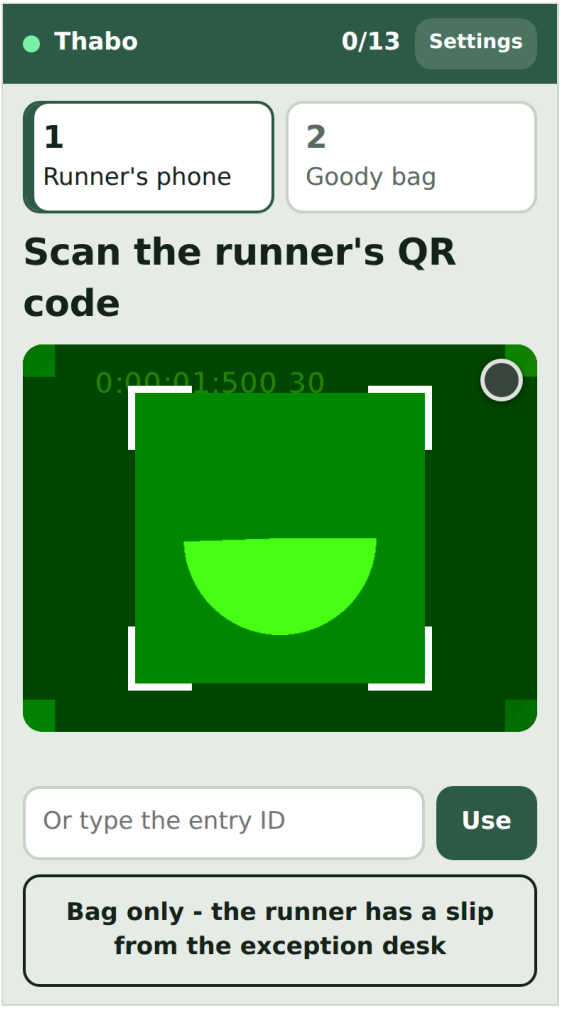


Figure 33. *Step one, waiting. The counter in the corner is the only number anybody at the table ever needs, and it climbs all evening.*

The 65 km runners start arriving at five, in cars loaded with tents and the complicated calm of people about to go and suffer on purpose.

The first is a teacher from Newcastle called Sibusiso Ngcobo, who holds out his phone with the email already open. Thabo points the camera at it.

The phone does not check him in. That is the change Thabo has had to unlearn since September, when a phone that saw a code acted on it, and a phone swung across a table full of labels on its way from the runner to the bag acted on whatever it passed. Now the camera only looks. When a code it can use is in the window, a round light in the corner of the picture turns green, the phone beeps once, and the tab at the top of the screen — *1 Runner's phone* — turns green and says whose code it is.

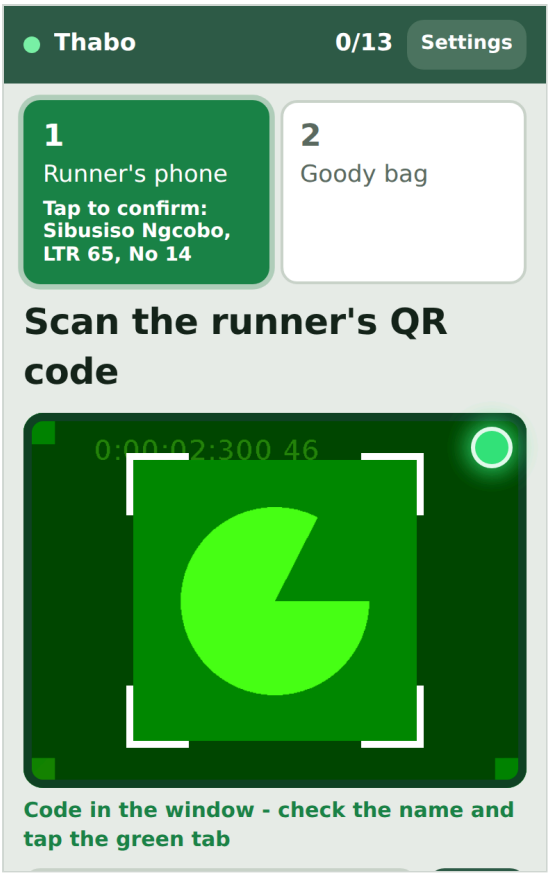


Figure 34. *The camera has seen a code, and nothing has happened yet. The light is green, the tab carries the name, and the next thing is a thumb.*

Thabo reads *Sibusiso Ngcobo, LTR 65, No 14* on the tab, looks at the man, and taps it.

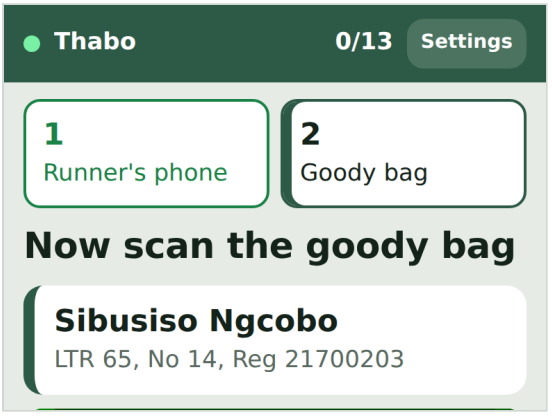


Figure 35. *The card comes up before the bag is touched: name, class, number. The volunteer now knows which bag to walk to, and step two is waiting.*

He finds bag 14 on the front table, second column along. On the way there the phone passes over bag 22, which is Jurie Oosthuizen's, and the light flicks amber and the second tab says so.

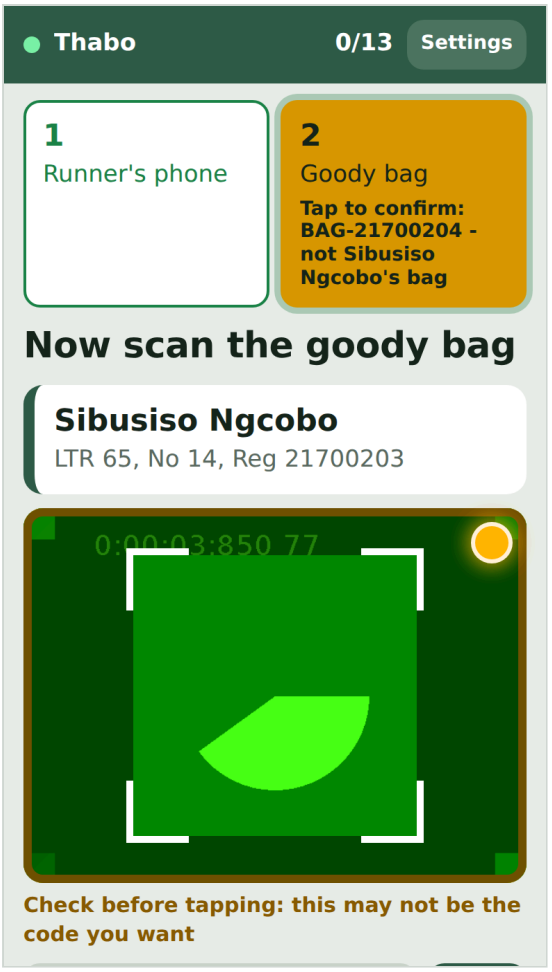


Figure 36. *Passing over the wrong label on the way. The light goes amber and the tab says whose bag it is and that it is not this runner's. Nothing has been sent to the laptop, and nothing will be unless somebody taps.*

"That," says Anke, over his shoulder, "is the whole point. Last year that would have been a wrong-bag screen and a beep and a queue watching you cancel it. Now it's a colour you ignore."

He holds the phone over 14. Green. He taps the tab, and the screen goes green all over.



Figure 37. *Green, with a beep loud enough for a marquee. Three buttons, and two of them are for things that have not happened yet.*

"Good luck up there," says Thabo, and means it. Anke ticks Ngcobo on the printed sheet on her clipboard, because the clipboard is the thing that does not run out of battery.

20. The Man With No Phone

At quarter to seven, Jurie Oosthuizen arrives. He farms the land on the northern side of the mountain and has therefore, in a sense, been training on the course for forty years. He has no phone. He has never had a phone. He has a bag of biltong, which he puts on the table as a matter of course.

Thabo, who has not yet met this particular exception in the wild, makes the mistake anybody would make and points the camera at the goody bag.

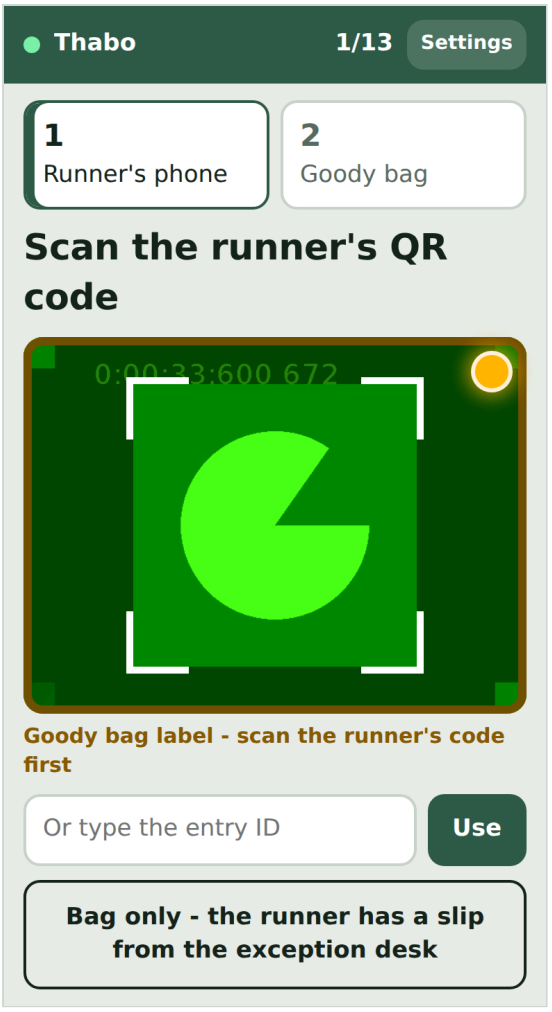


Figure 38. *A bag label at step one. The light goes amber and the line under the picture says what it is; the tab does not light, and there is nothing to tap. No class and no circumstance completes a handover from a bag alone — and the button at the bottom is where the rest of this chapter happens.*

"He's been to the desk," says Anke, over his shoulder. "Ask him."

He has. He is holding a strip of paper with OOSTHUIZEN, Jurie on it and one line under that: 22 Jurie Oosthuizen LTR 65. Thabo presses *Bag only — the runner has a slip from the exception desk*.



Figure 39. *The bag-only route does not ask for a bag. It asks for a race number off the paper, which is a thing only the desk can have given him.*

He types 22, the phone shows him one bag, he holds the camera over the label and taps the green tab, and the screen goes green:



Figure 40. *Handed over without anybody's code. The registration had one bag on it and now has none.*

In the record, somewhere Jurie will never see, that handover is marked *Email Scan Bypassed = Y*, and the audit log has the line: *checkin_bypass*, the bag, the volunteer, and *[slip 22]*.

Jurie takes his bag and nods, and leaves the biltong where it is.

21. What The Slip Will Not Do

This is the chapter Thabo wrote, on the Wednesday, by trying everything.

He presses *Bag only* and types a race number he has invented.

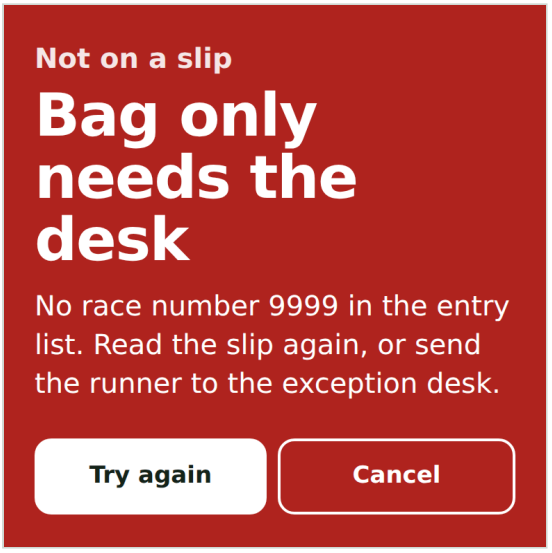


Figure 41. *A number that is on no slip and in no entry list. There is no partial credit here and no "check in anyway" button: the only way forward is the desk.*

Then he takes a real slip — Mthembu, two numbers on it — and types one of them.

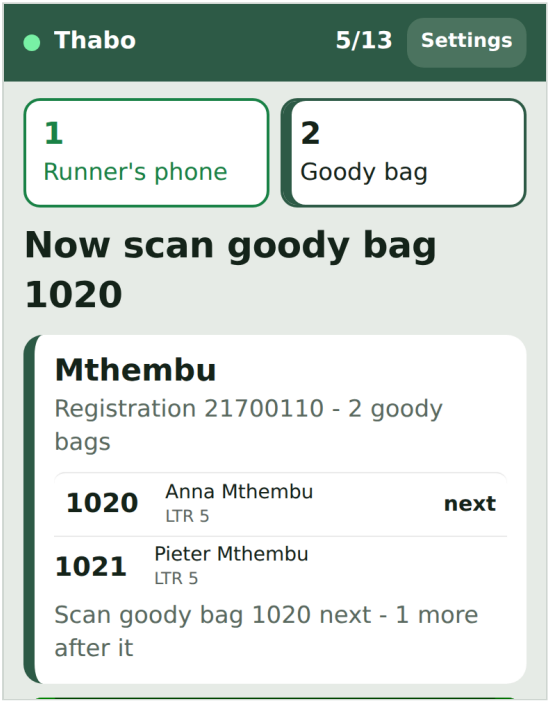


Figure 42. *One number off the paper opens the whole registration, exactly as the runner's own code would have. The paper has done the job the phone would have done.*

Last, holding that same slip, he holds the camera over a bag belonging to somebody else entirely. The tab turns amber before he has done anything:

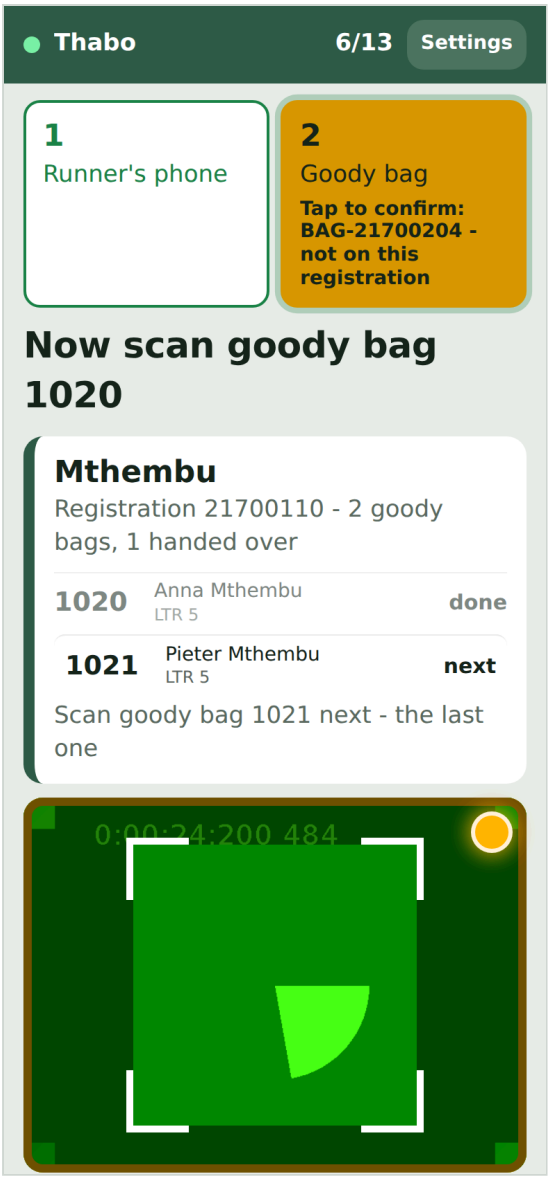


Figure 43. *The second tab, amber: that bag is not on this registration. Most volunteers stop here.*

He taps it anyway, because he is testing, and the phone refuses it by name.

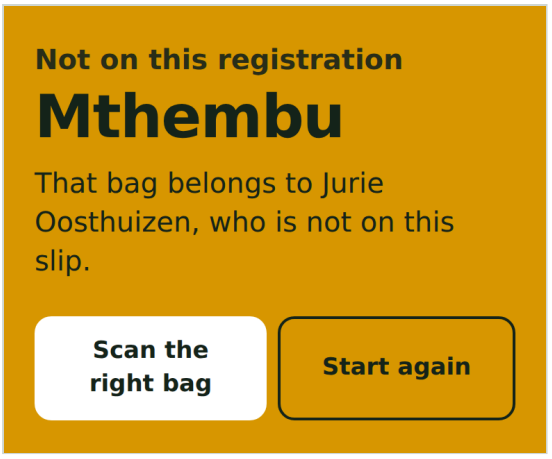


Figure 44. *Refused, and refused by name. The slip authorises those two bags and nothing else, and the phone says whose bag it is so the volunteer can put it back where it belongs.*

Then he does the thing the new rule is for. He takes Ruan Kruger's member number off the entry list, makes a QR code of it on his own phone — any phone will make a QR code of anything — and holds it up to the scanner.

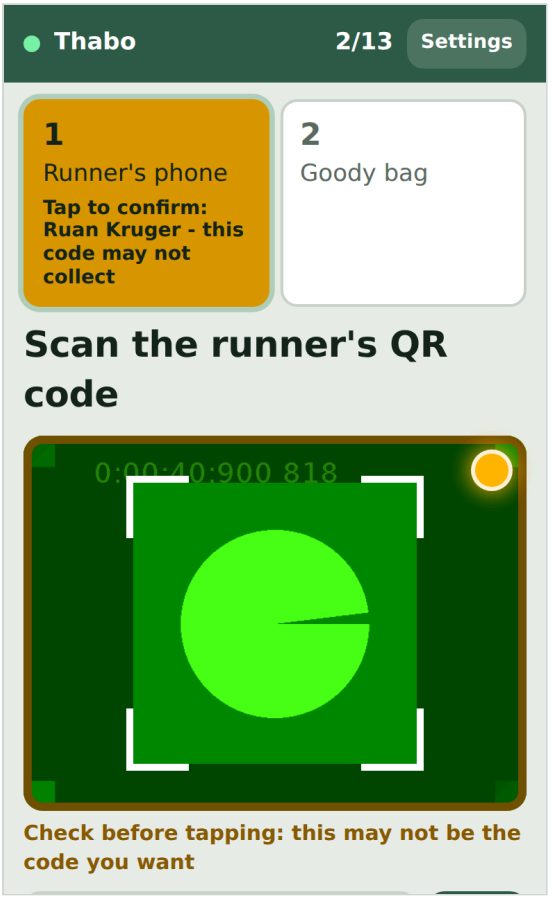


Figure 45. *The tab goes amber before anything is tapped, and says whose code it is and that it may not collect.*

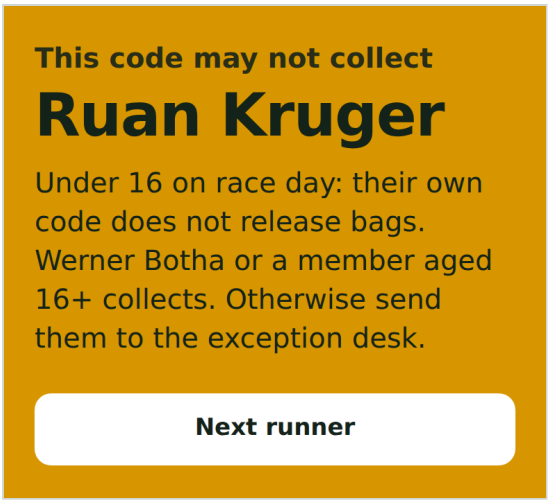


Figure 46. *Tapped anyway: the reason, and who collects instead. There is no button on this screen that hands anything over.*

"It's only a number," says Anke. "The rule isn't in the letter. It's in the laptop." Thabo sends the same number straight to the laptop from his own, without a phone in between, and the laptop says no there as well. Ruben's number, sent the same way, opens all three Kruger bags, because Werner said so in an email on Sunday night.

"So the paper is the lock," he says.

"The paper is the lock," says Anke. "The phone only checks that the bag matches the paper. Anyone holding a slip the desk tore off can take the bags on it, and nobody else's."

"And if someone just memorises a number?"

"Then a volunteer who is looking at them hands over a bag without seeing a slip, and that's a volunteer problem, not a software problem. There's a switch that makes the PC wait until the desk has logged the tear, if we ever want it. We don't, this year. A marquee where everyone knows everyone."

22. The Queue That Did Not Stop

Twenty minutes later, Marius Venter arrives with a tent still on his shoulder and four people behind him. Thabo holds the camera over his code and the tab goes amber rather than green: *Marius Venter, LTR 65, No 31*, and the line under the picture says to check before tapping.

He taps, and the whole screen goes amber.

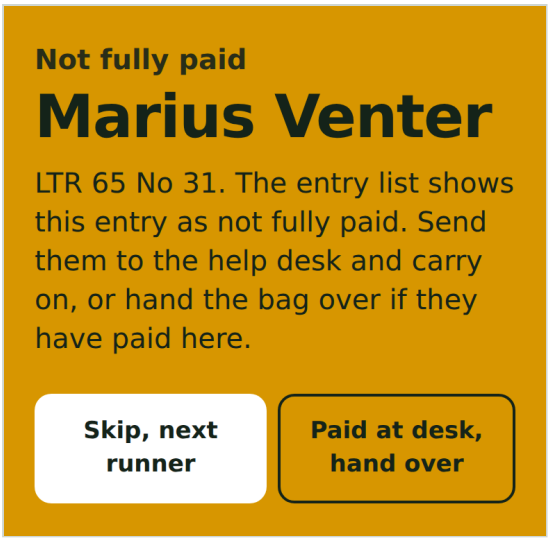


Figure 47. *Amber, not red, and look at the order of the buttons. Skipping is first and takes one thumb; handing the bag over anyway is second and deliberate.*

"It's the three hundred," says Marius. "I know. It's in the bakkie."

"Go and get it," says Thabo, and presses *Skip*, and the camera comes back, and the woman behind Marius is checked in eleven seconds later, and the queue does not stop.

Marius comes back four minutes later with three hundred rand, which goes into the cash tin and produces a receipt from the pastor, and Thabo scans him again, presses *Paid at desk, hand over*, and scans the bag.

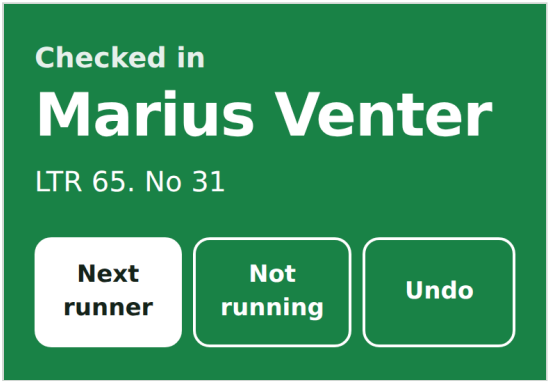


Figure 48. *The same handover, four minutes later, with the override recorded. The treasurer finds it on Sunday without anybody having to remember the conversation.*

"One person who hasn't paid," Anke says afterwards, "must never be allowed to stop the people behind them. Take his bag off the table if you have to. Just don't stop the table."

23. The Stick Falls Out

At eight, Oom Dawie reaches across the laptop to pass Anke a coffee, catches the red lanyard with his elbow, and pulls the USB stick clean out of its socket.

Nobody notices at first. The phone carries on scanning. Then Thabo glances at the laptop and the log pane has gone red and bold:

```
##### ALERT 08 #####
Backup drive is not writing: [Errno 20] Not a directory: 'usb/race_backup'
Check-ins continue and the PC's own files are complete.
Action: Reseat the stick or plug in another with the same drive letter. Mirroring resumes by itself. If it cannot be fixed, copy checkin_files by hand at the end.
See the setup guide, Alerts, number 08.
##### END ALERT 08 #####
```

At the same moment a red banner appears on the phone:

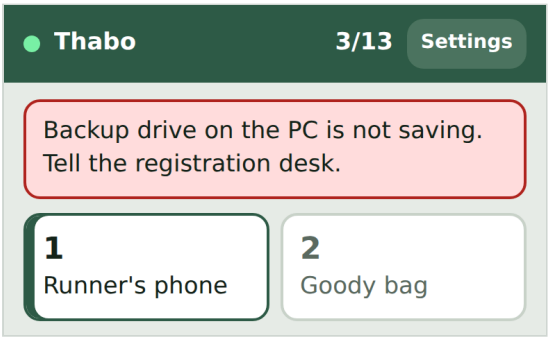


Figure 49. *The phone knows the laptop's spare copy has stopped, and says so in the only words a volunteer needs.*

"Tell the registration desk," says Thabo. "We are the registration desk."

Anke plugs it back in. The next scan writes a fresh copy of everything and the banner clears itself. Nothing was lost; the laptop's own record never stopped. The alert was only saying, as clearly as it could, the spare copy is not being made, and somebody should know.

Oom Dawie apologises four times. The lanyard is taped to the table leg for the rest of the evening.

24. The Wi-Fi

At about nine, when the braai fires start and somebody plugs a speaker into the extension lead, the Wi-Fi goes down.

Thabo only realises because the dot at the top of the phone turns red. He is halfway through checking in Lerato Fourie — the one entry of her, now — who has driven from Durban and looks very tired.

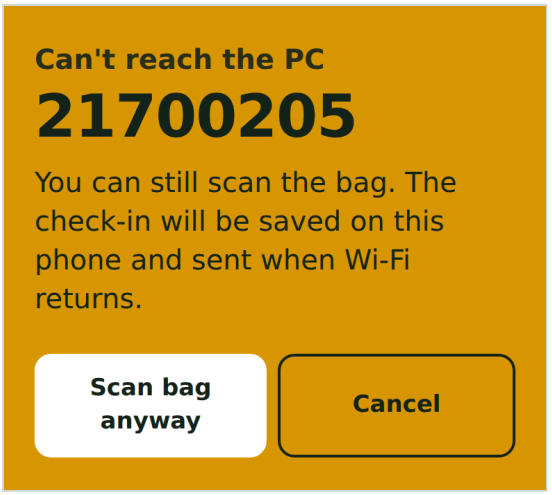


Figure 50. *The laptop cannot be reached, and the phone says what it can still do rather than what it cannot.*

He presses *Scan bag anyway*, holds the camera over her bag, and taps the green tab. The phone says *Saved on this phone*, in amber.

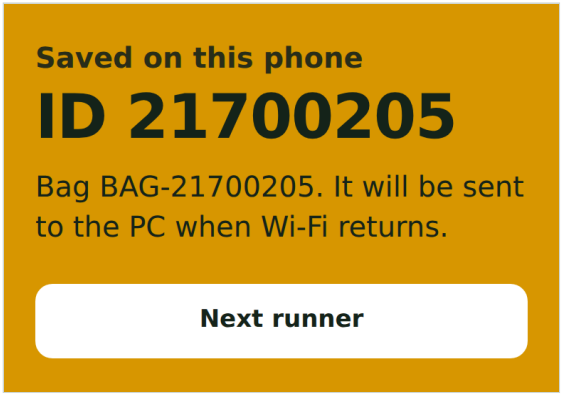


Figure 51. *The handover is kept on the phone, with the time it really happened.*

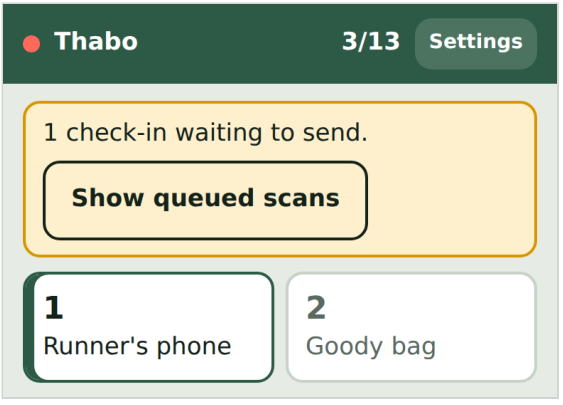


Figure 52. *And the banner on the scanning screen, counting what is waiting to go.*

Anke ticks Fourie on the clipboard. Lerato takes her bag and goes to find her tent. The rule is read aloud from the volunteer notes, because it is the one that matters: don't close the browser, don't swap phones.

The router comes back at twenty past nine, when Oom Dawie unplugs the speaker and the youth group protests. Within ten seconds the dot goes green and the check-in arrives on the laptop, carrying the time it really happened rather than the time it was sent.

Anke looks at her clipboard and is calmer than the phone is. If the router had never come back, the paper already said who had their bags.

25. The Phone That Died

The last Friday exception is the one Anke has always been most afraid of, and it arrives at twenty to ten in the hands of a volunteer called Nomsa, who has been scanning at the far end of the table on her own phone.

"It's dead. It went from fourteen percent to nothing."

At twenty-four minutes past nine, while the Wi-Fi was down, Nomsa checked in Mevrou Steyn from her printed invitation, and it is still sitting in a browser on a phone that will not turn on.

This is what the recovery flow is for, and Thabo has practised it twice. The phone goes on a charger for five minutes, long enough to come up — but not, it turns out, long enough to find the Wi-Fi again. Her browser is not closed — that is the rule that gets read out loud, and the reason it gets read out loud — so the queue is still in it. Nomsa taps *Show queued scans*, and her phone puts the handover on its screen as a code.



Figure 53. *A check-in that never reached the laptop, turned into a picture another phone can read.*

Thabo puts his own phone into *Recover scans from another phone* and holds it over hers.

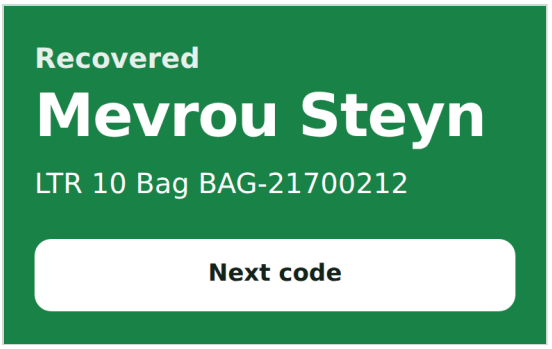


Figure 54. *One code, one scan. His phone says recovered and his count goes up by one.*

Then her phone asks, twice, whether every code on it has really been scanned, and will not clear the list until she says so.

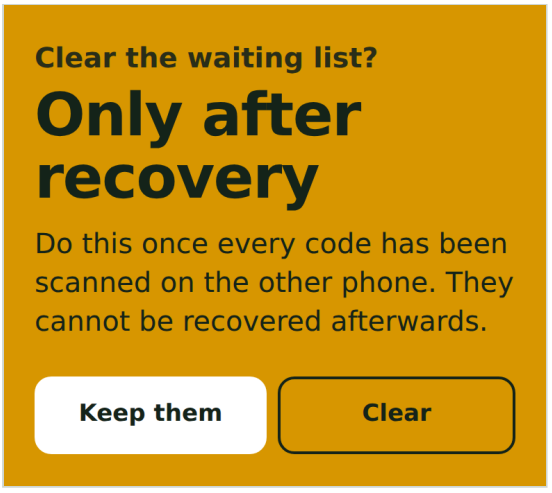


Figure 55. *Keeping them is the first button. Clearing is second, and cannot be undone.*

"And if she'd closed the browser?"

"Then it's gone," says Anke, "and we'd be reading the clipboard instead." She turns it round: Mevrou Steyn's name, ticked in pen at the time it happened.

26. Took The Bag, Not Running

At half past nine Marius Venter comes back to the table, the receipt from the pastor still in his hand, and stands there instead of leaving.

"I'm not going to run it," he says. "Knee went on Wednesday. I thought I'd walk it off, and I've just walked from the car and I can't. I'm here for the weekend anyway, and I've paid for the shirt, so."

Thabo scans him. The phone reminds him that Marius has collected, with the time and who handed the bag over, and underneath is a button that says exactly what Marius has just said.

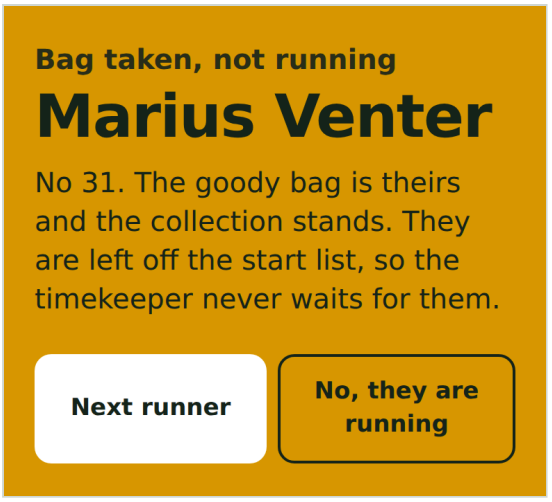


Figure 56. *The collection stands — the bag really is his and really is gone from the table — and the number comes off the start list in the same press.*

The check-in is untouched, but the row gets *Not Starting* = *Y*, and that one flag does exactly one thing. When the start list goes to RaceClocker, he is not in it, and the export says so by name rather than quietly:

```
1 entrant(s) took their goody bag and told the desk they are not running.
They are collected, and deliberately not on the start list:
       31  Venter, Marius (LTR 65)
```

"Why does that matter so much?" Thabo asks. "He'd just be a DNF."

"He'd be a name on Gerda's screen at six in the evening that hasn't come through the finish," says Anke. "And the sweeper's already down. That's not a DNF, that's a search."

27. Closing For The Night

At ten the marquee empties, and Anke presses *Export all*.

It asks first, because it stops the server, and on Friday night that is the point. The phones go quiet, the light on the wall goes red and says so in words, and four RaceClocker files are written from the evening's record — Friday's start list, for Gerda to look at over coffee.

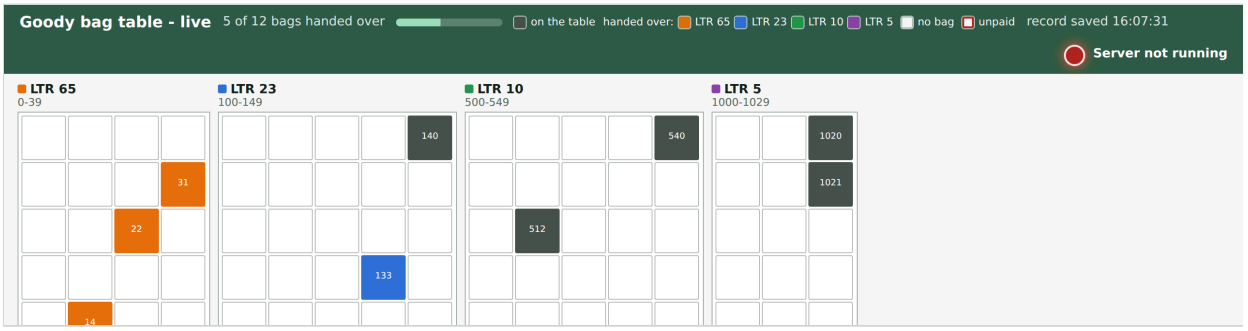


Figure 57. *The wall after Export all. The round light is red: the server is not running, and the table says so before anybody reaches for a phone.*

Last year, pressing Export was the end of the desk. The record was finished, and anybody who arrived on Saturday morning was a problem for a pen. The last line in the

log now says what to do about that:

```
Files are in output
Late runners? Press 3 to restart the server: scanning carries on and adds to the same record, so the
next Export has everyone.
```

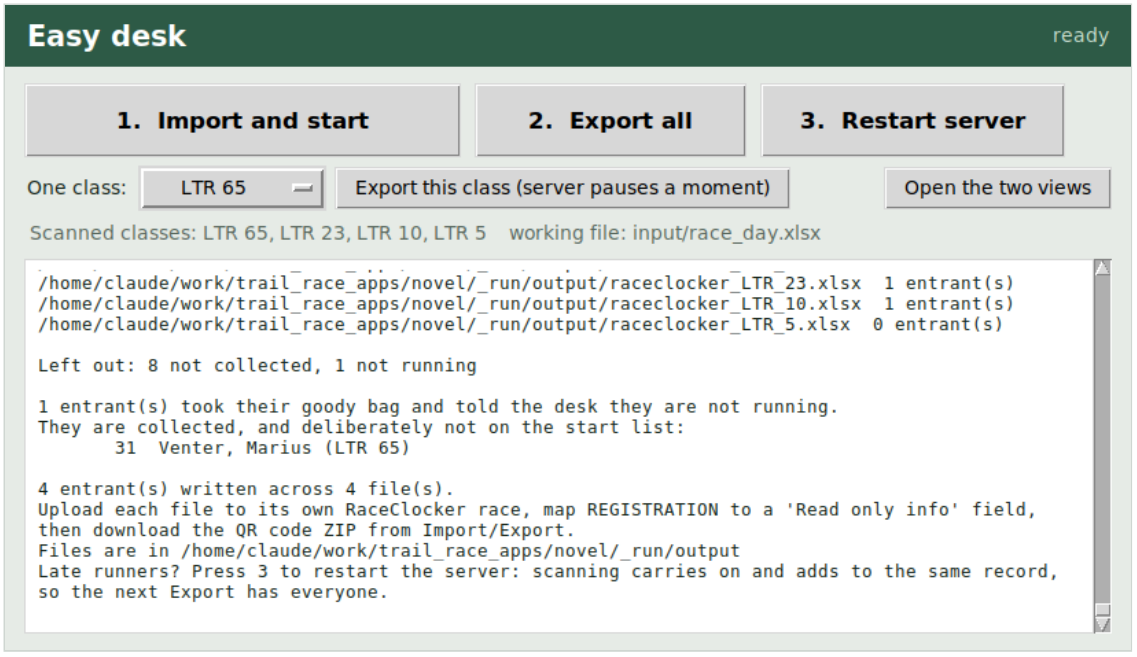


Figure 58. *The desk on Friday night, closed. Three buttons, and the third one is for the morning.*

28. Saturday, Before Light

At a quarter past four Anke presses *Restart server*. The laptop reads the record back and carries on where it stopped:

```
--- restarting the check-in server: it reads the record back and adds to it
Resuming from state.json (version 9)
13 entries loaded, 5 already checked in
Open on volunteer phones:
  https://lueneburg-trail-run.local:8443/
```

The phones, which were left open on the table overnight with their screens off, find it again by themselves. The light on the wall goes green.

At twenty to five Gerda comes to the desk for the 65 km start list, and the short races have barely started collecting. This is what the row under the three buttons is for. Anke picks *LTR 65* and presses *Export this class*. The server stops, the one file is written from the record, and the server starts again:

```
--- server paused for the LTR 65 export
output/raceclocker_LTR_65.xlsx  2 entrant(s)
--- server running again after 0.6 s
```

Well under a second. Thabo, who is holding a phone at the time, notices nothing at all, which is the correct amount. A bag scanned in that moment would have waited on the phone and gone a second later, like a Wi-Fi drop too short to see.

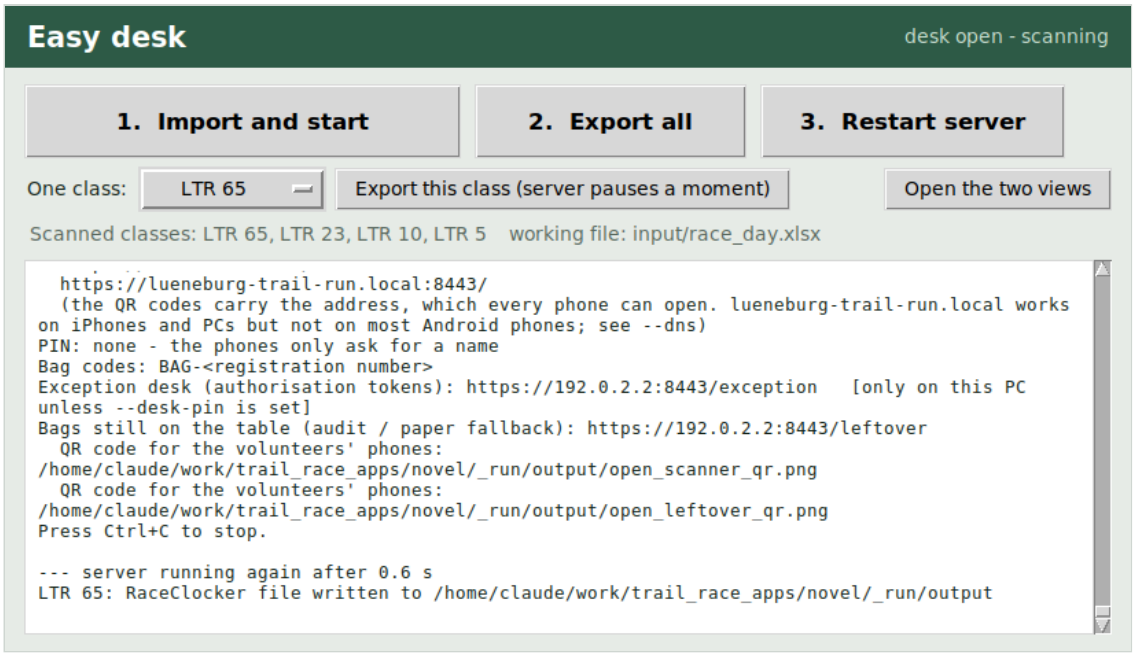


Figure 59. *The desk after Export this class: one file for one class, the server stopped and started again, and the time it was down written in the log.*

Twenty-seven collected. Twenty-six head torches at the gate.

The name missing from the marshal's clipboard is Andile Zulu, number 7. His tent is still pitched and his car is still in the field, and he is asleep in it with a temperature of thirty-nine and a note on the dashboard in shaky capitals: NOT RUNNING, SORRY, TELL ANKE.

This is the case the system cannot catch, and Anke says so plainly at the gate. The *not running* button works for the man who stands at the table and tells you. It does nothing at all for the man who collects on Friday evening feeling fine and wakes at three with a fever. Nothing in the laptop knows he is in that car.

What catches him is Gerda, who insisted months ago that the 65 km get a headcount at the gate as well, against the list of those who collected. Two lists, one made of scans and one made of a marshal counting torches, and the difference between them is a name.

Gerda takes Andile out of the RaceClocker start list before the first runner reaches the first checkpoint. He appears in the results as a non-starter rather than as a runner who vanished somewhere on a mountain. At a trail race that is not paperwork. It is the difference between a sweeper going home at dusk and a search team going up at midnight.

29. The Short Races

Six o'clock, and this is the morning the committee argued about in March.

Two hundred and thirty-five people in three races, all of them scanned, at a desk that used to run on a number and a name. Four volunteers, four phones, and the queue folded back on itself twice before seven. It is slower. Everybody had said it would be, and it is: about eighteen seconds a runner against the ten it used to take, and the tap on the tab has added a second to that which nobody minds, because it has taken away the wrong-bag screens that used to cost ten. Tannie Marietjie says one short sentence

about the queue at half past and then does not mention it again for the rest of the morning, which from her is a considerable concession.

What it buys arrives at twenty past six, in the person of Ouma Mthembu, who is seventy-nine, has come for the grandchildren, and has been to the exception desk.

She has the Mthembu slip — 1020 and 1021 — and the volunteer types the first number and gets both bags on the screen, which is figure 42. She hands over 1020.



Figure 60. *The first of two, on the desk's authority. The phone names the next number, so nobody has to hold it in their head.*

Then she reaches for what she thinks is 1021 and the tab goes amber, because the bag in her hand belongs to Jurie Oosthuizen, whose 65 km bag somebody had stood down on the 5 km table on Friday night. That is figure 43, and it is not a drill: it happened at twenty past six with a queue behind her, and the volunteer did not tap.

She finds 1021 two rows along and the tab goes green.



Figure 61. *Handed over without anybody's code, on the desk's authority, recorded as a bypass, with the slip already logged as torn. Four things had to line up, and all four are in the record.*

The Kruger registration comes through at twenty-five past, and is the reason the whole group flow exists: one code, three bags, two classes, two different tables.

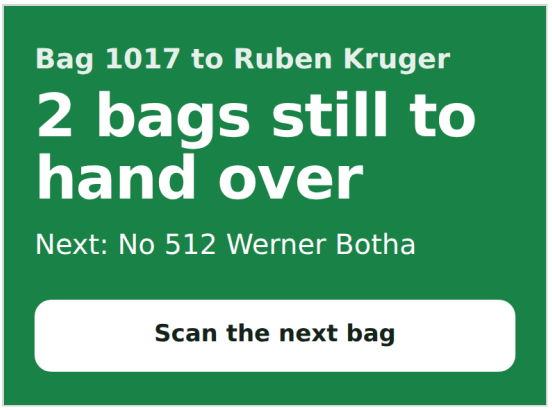


Figure 62. *After the first of three. The phone names the next number so the volunteer does not have to hold it in their head while walking thirty metres.*



Figure 63. *All three, on one scan of one code, in whatever order they came to hand.*

30. Four Small Wrongnesses

The bag that has already gone. At quarter to seven a man in a Luneburg rugby jersey holds out his phone, and the tab does not go green. It goes amber, with his name on it, and the words *already collected*.

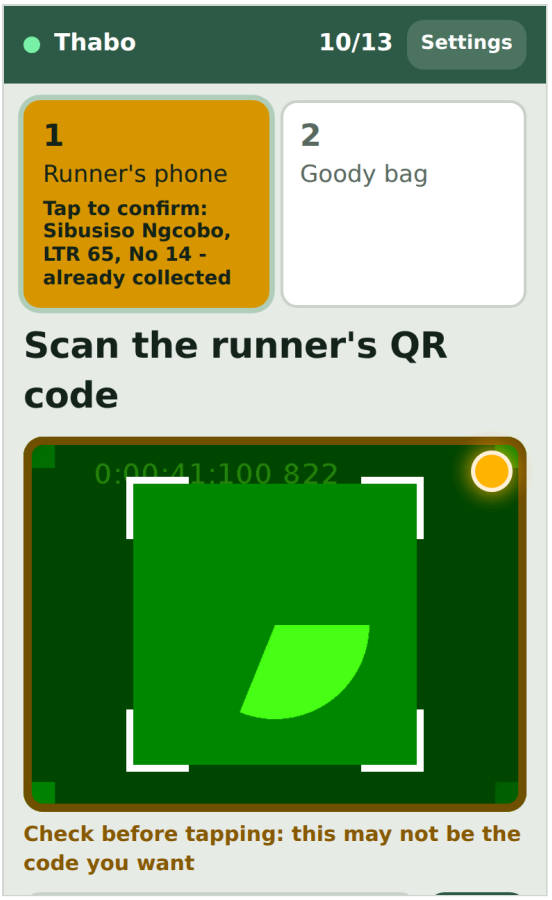


Figure 64. *Before anybody has tapped anything, the tab has said it.*

The volunteer taps it to see the rest.

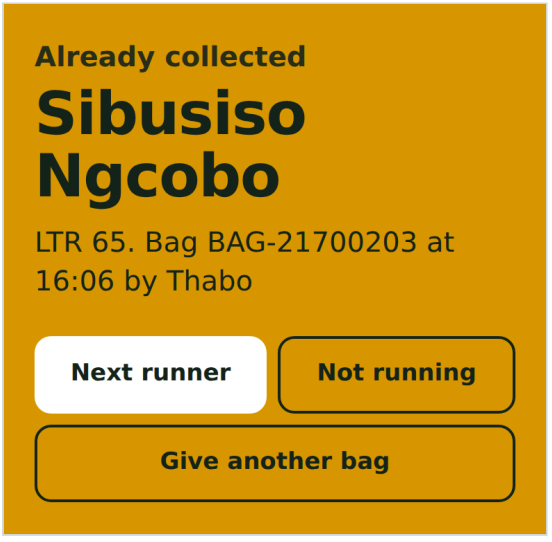


Figure 65. *Not an error — a fact, with a time on it and a name. Three seconds later he is on the telephone to his brother.*

His brother collected both their bags at half past six on his way to the toilets and forgot to mention it. Last year this conversation took four minutes and ended with a second goody bag being handed over to keep the peace.

The wrong bag. Just after half past six, a volunteer checks in Gerda van Wyk, reaches for 540 and comes up with the bag next to it. The Kruger cousins are not the only near-identical pair on that table. The tab has already gone amber:

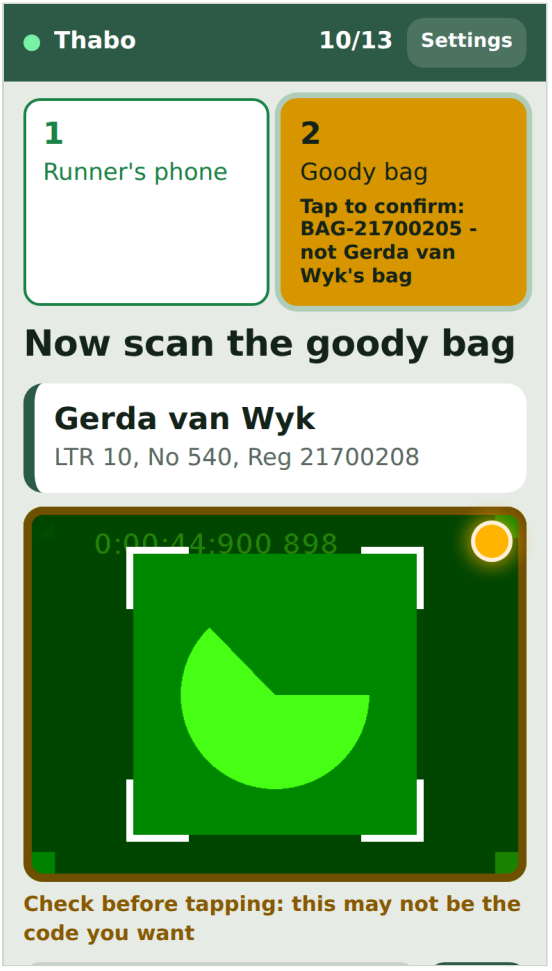


Figure 66. *The label and the runner disagree, and the tab says so while the bag is still in the volunteer's hand.*

She taps it to be sure.

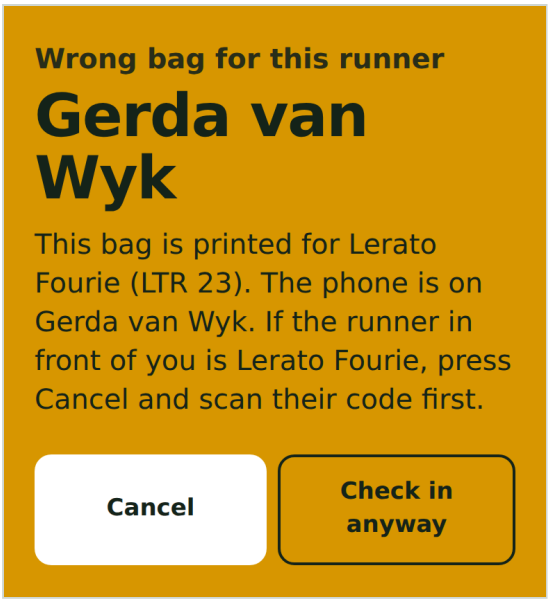


Figure 67. *And the phone says whose bag it actually is. Two people were about to go home with each other's shirts.*

The code that is not ours. At seven a man presents a QR code from a different race entirely, having opened the wrong email. The tab goes amber and says *not in the entry list*, and the volunteer taps it to show him.

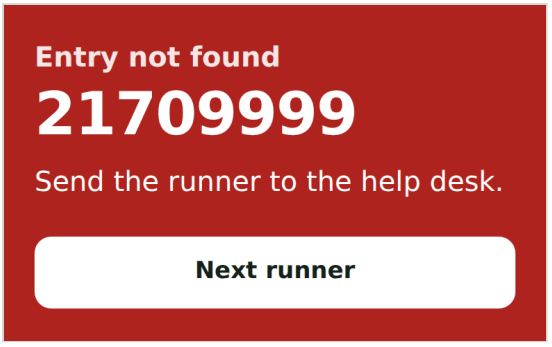


Figure 68. *Nothing in the list matches. No guessing, no partial match, and a sentence a volunteer can say out loud.*

The nudged, who came. Two of the eight arrive, which is two more than Johan expected. One is a woman from Utrecht at ten past six, holding her phone with the orange panel still on the screen: "It says I must come here first." The late entry desk is a trestle table with a card machine and a hand-written sign, and it takes four minutes: she pays, Anke gives her the next free number in the 10 km, writes it on a spare bag with a marker, and puts the bag in its gap.

The other is the one with no race number, who had meant to sort out his 23 km entry for a month and had not. He runs anyway, from the late entry desk, and says it is probably for the best.

The man Johan terminated does not come, which is the only part of the whole system that can never be proved right.

31. The Walk of the Table

At half past seven the marquee is quiet and the tables are mostly bare wood with bags here and there still on them.

Scanning never failed. So the walk of the table this morning is not the paper fallback — it is the audit.

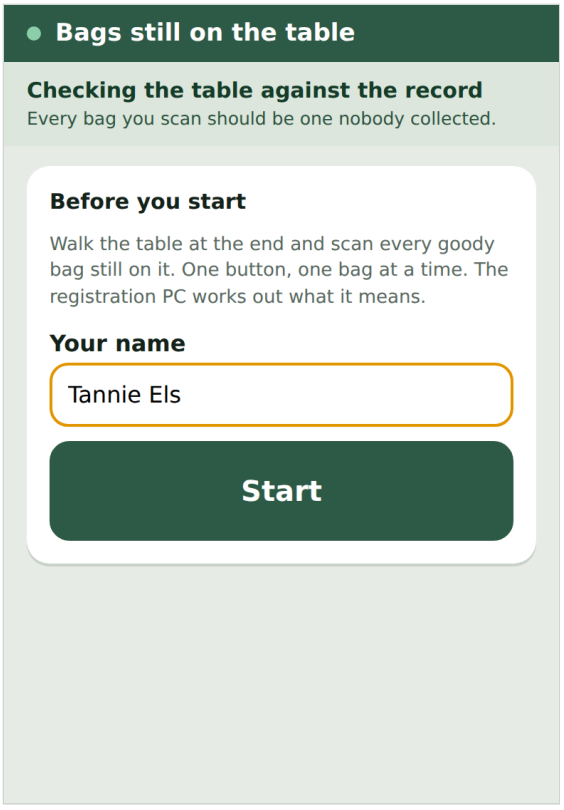


Figure 69. *The second app, opened for the first time all weekend. A name and one button, and it says on the front what the job is.*

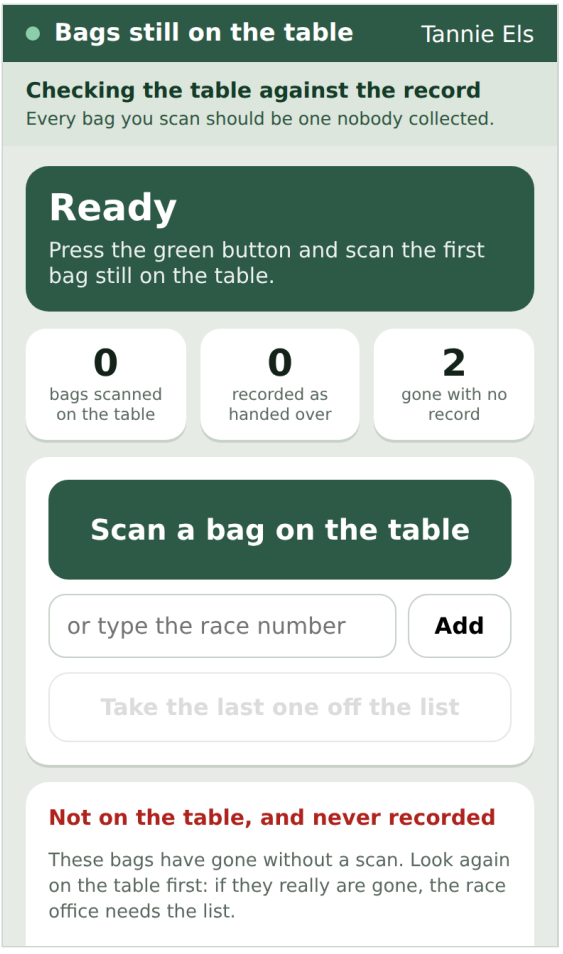


Figure 70. *Green banner. This walk changes nothing in the record. It only reports where the table and the record disagree.*

Tannie Marietjie walks the aisle with the phone in her hand, pressing one green button and scanning each bag still on the table.

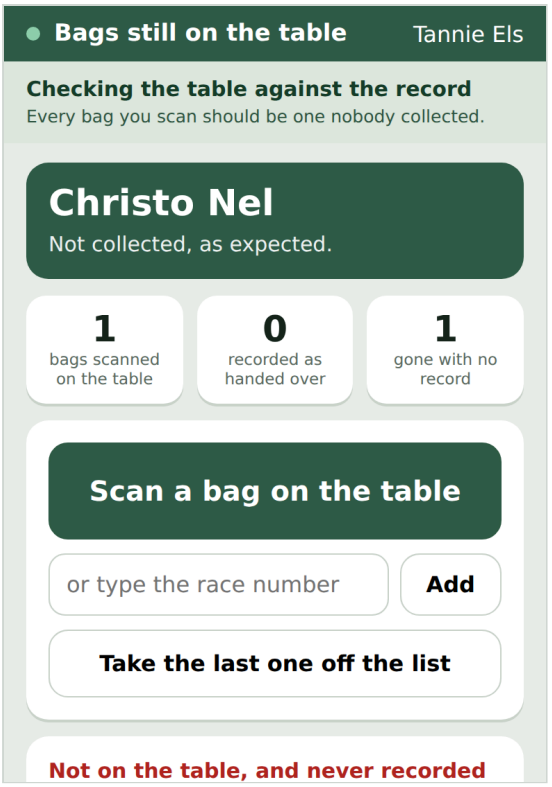


Figure 71. *Most of them look like this: a bag nobody collected, which the record also says nobody collected. No news is exactly what you want from most of a walk.*

Then, at the 65 km end where there should be nothing at all, there is one.

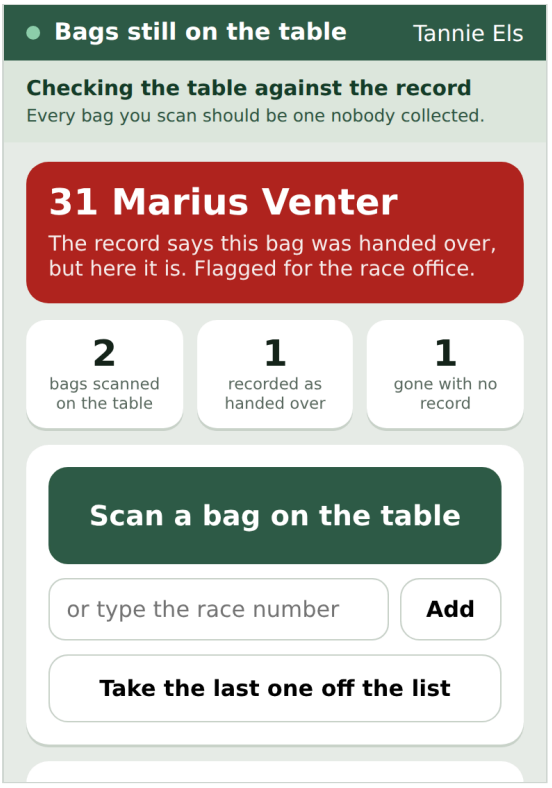


Figure 72. *Red, and the phone remembers who handed it over and when. The record says gone. The table says here. Both are true.*

It takes four minutes to work out, and is worth every one of them. Marius Venter — the man with the knee, the one Thabo marked not running on Friday night — took his bag, went back to his tent, decided on Saturday morning that he did not want a drawstring bag and a sachet of energy gel after all, and put it back on the table on his way to watch the start.

The record says collected, because it was. The table says here, because it is. The system has no way to know which came second, so it does the only honest thing and shows it to a person.

Tannie Marietjie writes *brought his bag back* on the clipboard, and the count goes on. When she finishes, the app shows the other half of the audit, which is the half nobody thinks of.

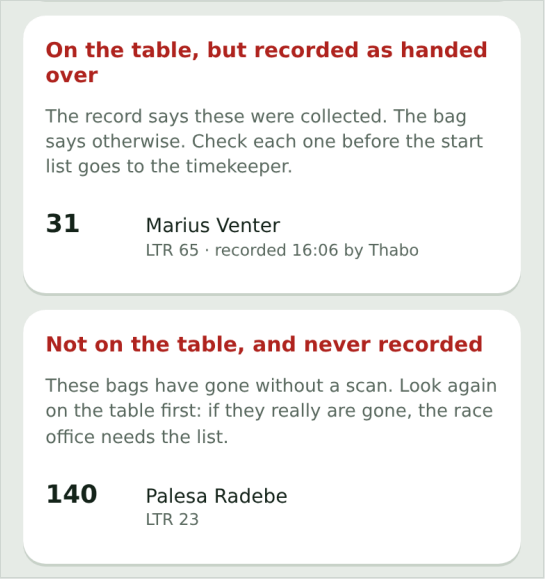


Figure 73. *Two lists. Bags here that the record says have gone, and bags gone that the record has never heard of. The second list is the one that finds a theft, in the year there is one.*

One name on that second list: Palesa Radebe. Her bag is found within ten minutes, on the table, one square out of place because somebody stood it down after reading its label. Nothing is missing. But the app asked the question, and in a year when something really does walk off, it will ask it the same way.

32. The Friend

At ten to eight, while the walk is still being argued over, Gerda van Wyk comes back into the marquee. She collected her own bag at half past six. She has come for Palesa's, and she goes to the far end first, because Palesa's letter said that was where the slip would be.

The pastor types *Radebe*.

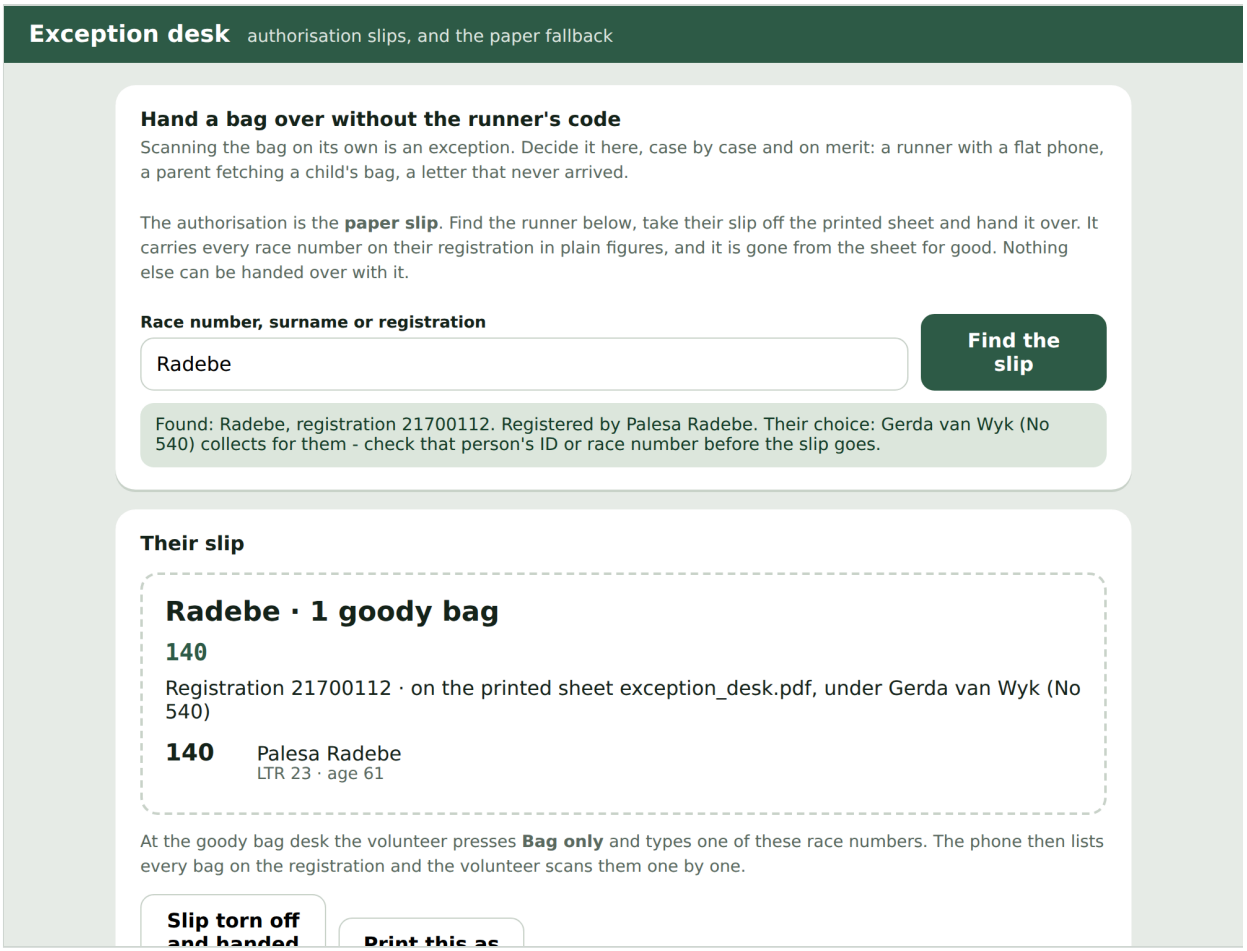


Figure 74. *The desk finds Palesa's registration and says, in the same line as the names, whom she chose to collect it — and that the slip is not on the 23 km sheet but on the exception desk's own, under the collector's name.*

Gerda shows her race number. The pastor finds the slip with her name across the top, already printed on Wednesday, tears it off and logs the tear. At the 23 km table the volunteer presses *Bag only*, types 140 off the slip, and holds the camera over the bag Tannie Marietjie has just put back in its square.



Figure 75. *Palesa's bag, handed to her friend on Palesa's own instruction, recorded as a bypass against the slip. Palesa is in Durban with a sprained ankle. Her T-shirt is not.*

33. What The Timekeeper Got

Anke makes the RaceClocker files for the last time at eight, from the day's check-in record — Friday's evening and Saturday's morning in one, because the server that restarted at a quarter past four added to the record rather than starting another. It takes ten minutes, including closing Excel when alert 15 tells her to.

	A	B	C	D	E	F	G	H
1	**Class**	**Entered**	**Checked in**	**Still to collect**	**Not fully paid**	**Bag-only**	**Bag taken, not running**	**On the start list**
2	LTR 10	3	3	0	0	0	0	3
3	LTR 23	3	2	1	0	1	0	2
4	LTR 5	4	4	0	0	2	0	4
5	LTR 65	3	3	0	1	1	1	2
6	Total	13	12	1	1	4	1	11
7								
8	Version	16						
9	Saved	2026-10-01 16:08:33						
10	Last change	checkin 21700209						

Figure 76. *The Summary sheet the server writes beside every version of the record. Entered, collected, still to collect, unpaid, bag-only, bag taken but not running — and the last column, which is the only one the timekeeper cares about.*

The last two columns are the ones that took a season to learn to keep apart. *Checked in* is a fact about a goody bag. *On the start list* is a claim about a human being on a mountain, and they are not the same number, and pretending they are is how a search party gets sent out at midnight.

There is a third thing the test found this year, before the weekend and not during it. A runner who pays at the desk is still *Fully Paid: No* in the entry list, because the entry list is EntryNinja's and the cash tin is not. Last year's export dropped anybody unpaid before it looked at whether they had collected — which meant that a man who paid three hundred rand at the table, took his bag and ran would have been on the mountain and on nobody's list. The export now asks the question the other way round: once collection has started, a bag handed over is a starter, and anybody who paid at the desk is named in the log as having done so.

```
1 entrant(s) not marked Fully Paid in the export collected their goody bag after paying at the desk,
and count as collected:
       31  Venter, Marius (LTR 65)
```

Four files. What is in them is the point:

- Everyone who actually collected a bag, including anybody who paid at the desk to do it.
- Not the ones who never came.
- Not the man who collected one and told the desk he was not running.
- Not Andile Zulu, who collected one and was asleep in his car with a fever, and whom the gate count caught rather than the software.
- Not the man Johan terminated, who was never in the file at all.

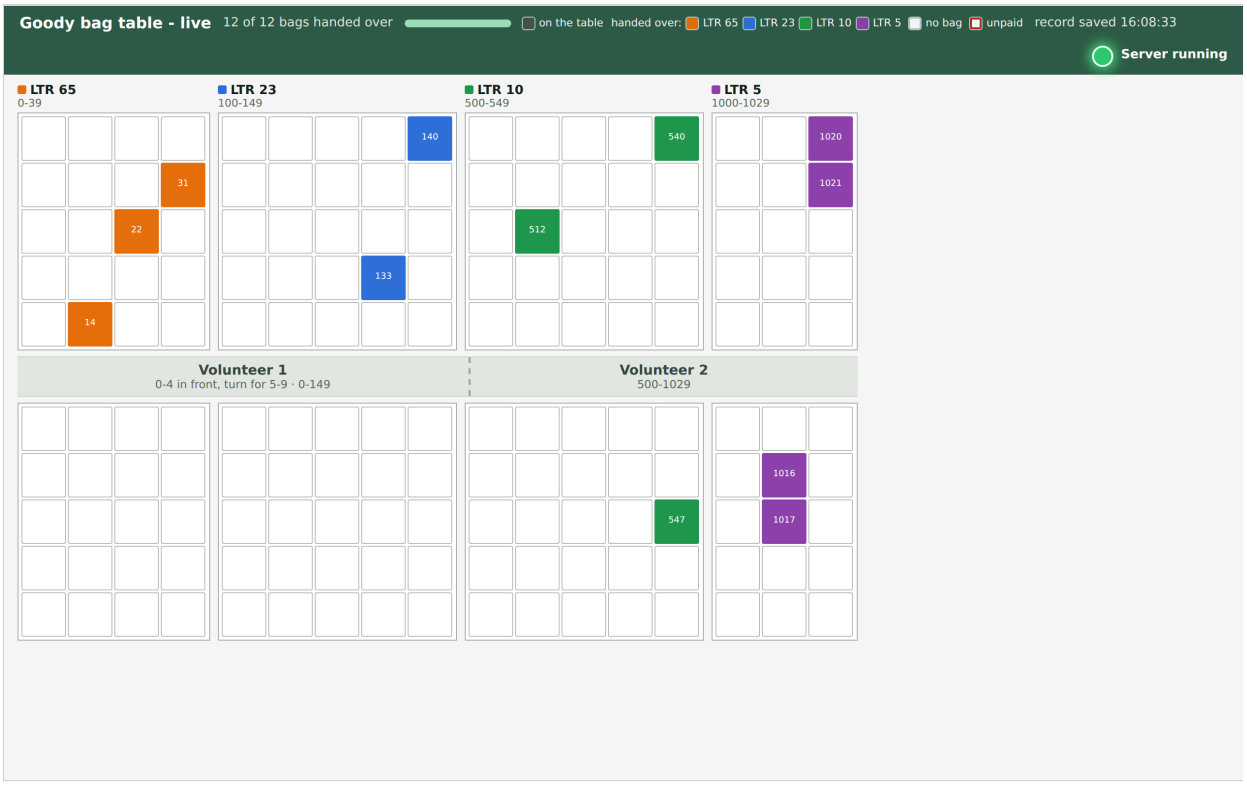


Figure 77. *The wall at the end. What is still grey is what is still on the table, and the two agree because the audit was run before anybody went home.*

Gerda uploads them as a replacement start list. When a number comes up on her screen at the finish, a person comes up with it, and there are no ghosts.

They need the camcorder twice — a 10 km runner whose number is folded double under a rain jacket, and a husband and wife who finish holding hands, as they have every race for eleven years. The video settles which of them crossed first: the wife, by a quarter of a stride, and she says so at the prize-giving, loudly.

At twenty past six, as the light goes orange over the mountain, the sweeper comes down with the last runner, a schoolteacher from Durban called Lindiwe Sithole, who walked the final four kilometres with a marshal because she had given everything she had on the climb.

Twenty-six went up. Twenty-six came down.

Anke sits in the cab of the bakkie for a few minutes where nobody can see her, and then goes back to fold the marquee.

Part Five: After

34. What The Exceptions Cost

The committee meets in the church hall on the Thursday, with the good urn.

Gerda goes first. "Last year it took forty minutes at dusk to be sure everyone was down. This year it took one look at the list. Two times needed the video. Both fixed. And I had the 65 km list at twenty to five, not at seven."

Tannie Marietjie says the desk was slower and she is not going to pretend otherwise. Eighteen seconds against ten. The queue reached the guy ropes twice.

"And?" says Johan.

"And nobody went home with the wrong bag," she says. "And the Kruger boy didn't have to stand there while I read four labels. And I didn't have to tell that man his brother had already been. And not one of my volunteers had to cancel a screen because the phone had looked at the wrong bag."

Then they count the exceptions, because that is what the weekend actually was.

what happened	how many	what caught it
duplicate race number	1	alert 07, before anything printed
excluded on import, named	5	the Excluded sheet
terminated by the desk	2	neither appeared
nudged, of whom two came and paid	8	the letter with a body
no race number yet	2	custom handling, kept off the start list
no email address	1	alert 18, her page printed instead
a child with a school email address	1	a letter with no code in it
replies to the collection letter, applied	2	the reply reader, then slimming again
replies left in the inbox, unread	3	alert 22; one settled by phone, one at the desk, one by silence
a message that was not a reply	1	not touched
a child's own code, made into a QR code by hand	1	amber tab; refused on the phone and on the laptop
a friend collecting with a pre-printed slip	1	the exception desk sheet
registrations spanning classes	7	one letter, one code, 19 bags
paper slips torn off	14	the gap in the sheet, and the desk's log
bag label in the camera at step one	9	amber light, nothing sent; the slip button
other labels passed while panning	dozens	amber light, nothing sent until a tap
bag-only attempts with no slip	3	refused outright
bag offered against the wrong slip	1	amber tab, then refused by name
unpaid at the table	4	skipped, three came back and paid
paid at the desk, then ran	3	on the start list, named in the export

what happened	how many	what caught it
USB stick out	1	alert 08, and a banner on every phone
Wi-Fi down	11 minutes	check-ins queued and sent
a phone that died with a scan on it	1	recovered off its screen
took the bag, not running	1	off the start list, named in the export
collected, then fell ill	1	the gate count, not the software
one class wanted before the rest	1	Export this class, under a second
late runners after the export	11	Restart server, same record
wrong bag reached for	1	amber tab, then the label and the phone
the next runner scanned before the last one's bag	6	tab 1 offers to switch; nothing sent
double collection attempt	1	refused, with a time and a name
a code from another race	1	not found, no guessing
bag brought back after collection	1	the audit, in red
bags a square out of place	1	the audit's second half
a file open in Excel	2	alert 15
the table rehearsed on the living-room floor	120 scans	the load-test paper: runners 33 to 92, MAIL and GOODY BAG

"Thirty-five kinds of thing going wrong," says Johan, reading it, "and not one of them lost a bag or a runner."

"Thirty-five kinds of thing that were going to happen anyway," says Anke. "They happen every year. The difference is that this year each of them had somewhere to go."

Thabo says the only thing he would change is to put the laptop further from Oom Dawie's elbow.

Johan asks whether they should go back to number-and-name for the 5 km, which is mostly children and mostly local.

"Ask me again when the 5 km stops being a hundred people I've never met," says Tannie Marietjie, who two months ago argued the other way, and that is the end of it.

35. The Envelope

That night, clearing the kitchen table for the first time since August, Anke finds the envelope. It has gone soft at the corners and there is a tea ring on it, and the list has grown up the side and round onto the front in four colours of pen.

> What does the desk actually need? A name. A distance. A number. Whether they've paid.
>
> An emergency number, for the medics. Underlined twice.
>
> A stopped run is better than a shortened list.
>
> Input goes in, output comes out, and only one of them may be thrown away.

One registration, one letter, one code, however many bags.

Sixteen and over collects for the group. A child's code opens nothing.

Let the program read the replies, and let it leave alone the ones it cannot read.

Terminate or nudge. Never just leave it and hope.

Scan the runner, then scan the bag. Every class, every time.

The camera looks. The thumb decides.

A bag on its own is an exception, and an exception arrives on paper.

The paper is the lock. The phone only checks the bag against the paper.

A practice round is a real round.

One person who hasn't paid must not stop the queue. One who paid at the desk must not vanish from the list.

Say which file is open, not which error it was.

Green is an audit. Red is the record. Only the desk may change it.

Collected is not the same as started.

An export is not the end of the desk. Restart, and keep adding.

Give the laptop a name. A number means nothing to a volunteer.

Count the leftovers before the urn goes back.

And across the bottom, in capitals, pressed so hard the pen has gone through:

NEVER CLOSE THE GAPS.

She pins it to the fridge, under the magnet from the Lutheran bazaar, where it stays until the following August, when the export arrives again, and she takes it down, and reads it, and opens the laptop, and starts.

The end.

Appendix A: the workflows, and where each exception goes

For whoever takes the desk over. Not how any of it was decided — only what it does, and what happens when it goes wrong.

The folders

folder	what is in it
input	the EntryNinja export, the slimmed race_day.xlsx written beside it, and the file of decisions about members who do not comply. Read constantly, cleaned never.
output	everything the apps make. Clean system empties this and nothing else.
racedesk	the window and the easy desk, and what they remember.
novel	this book: its text, the screen test that photographs it, and the builder.

The four races

race	numbers	collection	letters
LTR 65	0–99	scanned: runner's code, then the bag	Word letters, merged with a picture card each
LTR 23	100–499	scanned: runner's code, then the bag	one letter, one code
LTR 10	500–999	scanned: runner's code, then the bag	one letter, one code
LTR 5	1000–1499	scanned: runner's code, then the bag	one letter, one code

There is one collection mode and it applies to every class. The runner's code first, the goody bag second, and nothing completes from one of those alone. A registration covering several entrants is one code and several bags, in any order, across any classes.

The phone. One field: the volunteer's name. No PIN, unless the server is started with one, in which case the PIN field appears by itself. The QR codes carry the laptop's IP address, which every phone opens. The laptop also answers to *lueneburg-trail-run.local* (iPhones and PCs; not most Android phones), and — when the router hands the laptop out as its DNS server and the server runs with *--dns* — to *https://lueneburg-trail-run:8443/* on every phone, which then goes on the codes instead. The camera never acts on its own: a usable code in the window turns the light green and the active tab green with a name on it, and a tap on the tab is what goes ahead. A code of the wrong kind turns the light amber and is ignored; a code of the right kind that looks wrong — someone else's bag, an unpaid entry, an entrant already collected, a code not in the list — turns the tab amber, still tappable, so the volunteer sees it first. A code that has left the window stays tappable for four seconds.

The easy desk. *1. Import and start* does everything up to an open desk. *2. Export all* stops the server and writes the RaceClocker files. *3. Restart server* starts it again exactly as it was, reading the record back, so late runners are added to the same

record and the next Export has everyone. *Export this class* writes one class's file with the desk open: the server pauses for under a second and starts again by itself. The live table on the wall is the tables from above: a bag still on the table is dark grey, a bag handed over is its race's colour — orange 65 km, blue 23 km, green 10 km, purple 5 km — and its round light is green while the server runs.

The exception process, which is the only thing that is not that:

- The runner cannot show a code. They go to the exception desk.
- The desk finds them by race number, surname or registration number — this is where number-and-name still lives, and it is a search, not a handover.
- The desk tears their slip off the printed sheet and logs the tear.
- At the table the volunteer presses *Bag only*, types a race number off the slip, and the phone lists every bag on that registration.
- Each bag is scanned, confirmed with a tap, and recorded as a bypass, with the slip number in the audit log.

Nothing else hands a bag over without the runner's code. There is no per-class setting that turns this on, and no button that skips the tear.

Who may collect. The person who registered, or anyone on the registration aged 16 or over on race day, collects every bag on it with their own code; an under-16's own code is refused, on the phone and on the laptop. The last letter, sent the Sunday before, lets the person who registered change that once by replying 1 (a named under-16 may collect), 2 (only they may) or 3 (an entrant outside the registration collects, on a slip printed with their name). *Read reply emails now* applies the clear replies and leaves the rest unread with a reason (alert 22); *Collection choices* is where a person enters those by hand. Slim again afterwards. The exception desk sheet — delegate slips, the choices, the codes that do not collect, the replies still to sort — is printed by every slim with the rule on and every *Import and start*.

The exceptions, and where each goes

when this happens	this catches it
two entrants share the identifying value	the run stops before anything prints
a race number is used twice	alert 07: excluded, both named, on the Excluded sheet
a record has no class, no number, or is unpaid	excluded by name, on screen and on the sheet
an entrant will not pay or answer	terminate at the desk: off the list, named on the sheet
an entrant might still pay	nudge: imported anyway, one letter with a body saying what is outstanding
a race number is issued too late	imported with custom handling, kept off the start list until the desk deals with them
a registration has no email address	alert 18, and their invitation page is printed instead
a child under 16 shows their own code	amber tab and refused, on the phone and on the laptop, with who collects instead
a family wants a different rule	one reply before the deadline: 1, 2 or 3; the reply reader applies the clear ones

when this happens	this catches it
a reply cannot be read clearly, or came late	left unread in the inbox, alert 22; entered by hand on Collection choices, or settled at the desk
a friend outside the registration collects	option 3: a slip printed with their name on the exception desk sheet, off the class sheets
one registration covers several entrants, in any classes	the slimming tick box decides: ignored (the default) gives each their own letter, code and bag; kept gives one letter, one code and every bag on it, scanned in any order
the camera passes over other labels while panning	only the light moves; nothing is sent until a tab is tapped
a runner's code comes up while the phone still holds the last runner	tab 1 turns amber and offers to switch; the last runner's bag was never scanned
a goody bag label is in the camera at step one	amber light, nothing sent; the runner goes to the exception desk
a runner cannot show a code	a slip torn from the printed sheet: Bag only, then a race number off the slip
a bag is offered against a slip it is not on	amber tab; if tapped, refused by name, with the real owner named
a race number is typed that is on no slip	refused: there is no way past this but the desk
a runner has already collected	amber tab; if tapped, refused with the time and the volunteer
the bag does not match the runner	amber tab; if tapped, refused with the name the bag is printed for
an entry is not fully paid at the table	amber tab; then Skip, next runner first, Paid at desk, hand over second
somebody paid at the desk	on the start list: the export counts a collected bag as a starter, and names them
an Android phone cannot find the .local name	the QR code carries the number; with the router pointed at the laptop, every phone finds the plain name
the USB stick comes out	alert 08 on the PC, red banner on every phone; check-ins carry on
the Wi-Fi drops	the phone queues the scan and sends it when the router returns, with the time it really happened
a phone with queued scans dies	recover them from its screen onto another phone, before its browser is closed
somebody takes their bag and is not running	Not running: collection stands, omitted from the start list, named in the export
somebody collects and then falls ill overnight	nothing in software. The head count at the gate
one class's start list is needed early	Export this class: the server pauses under a second
runners arrive after the export	Restart server: the same record carries on
a file is open in Excel	alert 15, naming the file

when this happens	this catches it
the table has to be rehearsed before there are any bags	after the first 32 on the proof page, Import and start's load-test paper carries on: runners 33 to 92, 12 to a page, MAIL pages then GOODY BAG pages
a practice round blocks the next one	stop the server and clear the check-in record. Never on race day
scanning fails outright	the desk declares it. Only then does the table walk become the record
scanning works	the table walk is an audit: a bag here that the record says has gone, and a bag gone with no record, are both reported and neither is written

The two phone apps, whose QR codes are written into output when the server starts: *open_scanner_qr.png* for the table, *open_leftover_qr.png* for the walk at the end. Print both and tape them down.

The three things that stop everything: two people sharing one identity, one race number on two entrants in the identifying column, and a class larger than the table it was given.

Appendix B: what the one rule costs, and when to revisit it

Kept out of the story because it is about a future race, not this one.

Scanning never buys speed. It can only buy correctness. A volunteer who knows the runner hands a bag over in about seven seconds; one who does not takes about ten, because they must read the label and the name, which is the only check there is when a face means nothing. Two scans take about eighteen to twenty, whoever it is. The confirming tap adds about a second, and takes away the cancelled wrong-bag screens that cost ten each when they happened.

So the question is only how often the older way got it wrong. Fitted to what a weekend actually shows — one wrong bag reached for and one double collection attempt in two hundred and fifty-six handovers, both caught — and to which way each would have gone among strangers:

per handover	known runner	stranger	scanned
bag that actually goes astray	about 1 in 660	about 1 in 80	about 1 in 2000

Those are estimates, and they are the weakest page in this book. Worked four different ways, with four different sets of guesses about what a lost goody bag costs, the crossover lands between 55 and 75 per cent strangers. Luneburg's 23 km and 10 km crossed two-thirds this year; the 5 km has not, and is scanned anyway, because one marquee with two rules in it is a worse idea than either rule.

Going early has a price. Putting a race on the phone at one stranger in five is estimated to cost three to fourteen entries the following year — people who liked the farm and did not like the queue — against roughly one saved by the goody bags it protects.

There is a middle way this race used to run by accident and may run again on purpose: scan only the runners the volunteers do not recognise. That buys most of the correctness for a third of the delay, and it puts the check exactly where the knowing stops. It costs one thing, and it is not a small thing: two rules in one marquee, and a volunteer deciding which applies while somebody watches them decide.

What none of this measures is the warmth. The model can only say how many goody bags that warmth is allowed to cost before it stops being a bargain. At Luneburg, on these numbers, the answer is between one and two a year.

Appendix C: the test log

Every workflow and every exception in this book was run by one test, *novel/make_screens.py*, against a live check-in server and the real easy desk, on an entry list of thirteen entrants built to contain all of them: a registration of three across two classes, a registration of two in one class, an unpaid entry, an entrant with no race number, an entrant with no email address, a race number issued twice, three children of twelve, thirteen and fifteen, and six replies to the collection letter. The test plays the story in order — the file, the replies, the printing, the desk on Wednesday, the drill, Friday evening, the night, Saturday morning, the walk and the export — and takes every picture in the book as it goes. Run it again and the book is photographed again.

The one thing the test stands in for is the camera's decoder. Where a phone "sees" a code, the test hands the decoded text to the same function the camera calls when it reads one; everything after that — the light, the tab, the tap, the server — is the software itself. Times on the screens are the times the test ran. The folder the test worked in is left off the front of file names in the quoted logs. Alert 15 is quoted rather than re-run, because it needs Windows holding a file open, and the slip counts in chapter 16 come from the full Luneburg export rather than the test list.

Each row is what was tried, what the server or the program answered, and where in this book the screen appears. Rows marked *new* test what changed in this edition.

#	what was tried	answer	figure
1	read the replies to the collection letter new	2 applied, 3 left unread in the inbox, the parking question not touched	
2	reply: kruger.family@example.com new	apply — registration 21700900 (Werner Botha): 1 - Ruben Kruger (under 16) may collect for the group	
3	reply: sibusiso@example.com new	apply — registration 21700105 (Sibusiso Ngcobo): 0 - standard rule	
4	reply: lerato@example.com new	leave — option 3, but no race number or name of another entrant found	
5	reply: palesa@example.com new	leave — option 3, but no race number or name of another entrant found	
6	reply: marius@example.com new	leave — arrived Thu 08 Oct 08:15, after the reply deadline	
7	a reply left in the inbox, entered by hand on the Collection choices screen new	Saved: registration 21700112 (Palesa Radebe): 3 - Gerda van Wyk (No 540) collects; slip waits at the exception desk. Slim again to apply it.	15
8	Import and start makes the exception desk sheet, unasked new	1 delegate slip(s), 2 choice(s), 2 code(s) that do not collect, 2 reply(ies) still to sort, 1 with no letter at all, 2 on somebody else's letter, 5 page(s)	16
9	Import and start, 13-entrant list: the load-test paper new	No load-test paper: the proof page has 10 runner(s), so there is nothing from number 33 on (the screen covers them all).	
10	the load-test paper for a 259-entrant field new	runners 33-92 of 259 in the proof page's order, 5 MAIL page(s) then 5 GOODY BAG page(s), 12 to a page	21

#	what was tried	answer	figure
11	the scanner app's QR code, without --dns new	carries the IP address; lueneburg-trail-run.local printed under it	4
12	the letter's attached PDF: how collection works and the choices new	Goody_bag_collection_options.pdf	10
13	ask the Wi-Fi where lueneburg-trail-run.local is (mDNS) new	answer — 192.0.2.2	
14	--dns: the plain name, as an Android phone asks for it new	answer — lueneburg-trail-run -> 192.0.2.2; lueneburg-trail-run.lan -> 192.0.2.2; lueneburg-trail-run.fritz.box -> 192.0.2.2	
15	the certificate carries the name new	lueneburg-trail-run.local, lueneburg-trail-run	
16	a phone with no PIN asks for the numbers new	200 — 13 entered	
17	find a registration by surname at the desk	200 — Mthembu, 1020, 1021	27
18	the desk logs the slip as torn off	200 — registration 21700110	28
19	practice round: the same runner scanned again	409 — already	
20	the output folder is deleted under a running server	200 — made again, check-in written	
21	the phone opens with a name and no PIN new	200 — scanning	32
22	the camera sees a runner's code new	green light, tab says: Sibusiso Ngcobo, LTR 65, No 14	34
23	look up a runner by their code	200 — Sibusiso Ngcobo	34
24	panning past another runner's bag at step two new	amber, tab says: BAG-21700204 - not Sibusiso Ngcobo's bag	36
25	...and what went to the PC while the phone was panned new	0 check-ins sent	
26	a code that left the window, after four seconds new	dropped	
27	hand the bag over (tap the green tab)	200 — checked in	37
28	the next runner's code while the phone is still on the last one new	tab 1 amber: new runner Gerda van Wyk - the bag for Palesa Radebe was not scanned; nothing sent	20
29	a goody bag label in the window at step one new	amber: Goody bag label - scan the runner's code first; nothing sent	38
30	a goody bag label sent at step one (typed)	400 — scan_runner	
31	bag only, with the slip	200 — bypass recorded	40
32	bag only, with no slip	403 — no_slip	
33	a race number that is on no slip	404 — not_found	41
34	a QR code of an under-16's own number, in the window new	amber, tab says: Ruan Kruger - this code may not collect	45

#	what was tried	answer	figure
35	...tapped: an under-16's own code new	403 — Under 16 on race day: their own code does not release bags. Werner Botha or a member aged 16+ collects.	46
36	the same code sent straight to the PC, past the phone new	403 — may_not_collect	
37	Ruben, 13, whom his registrant named (option 1) new	200 — opens the registration: 3 bags	
38	Pieter, 15, on a registration with no choice new	403 — may_not_collect	
39	an entry that is not fully paid	200 — paid: no	47
40	paid at the desk, hand over	200 — checked in with override	48
41	the USB stick is pulled out	alert 08 on the PC; phone: Backup drive on the PC is not saving. Tell the registration desk.	49
42	took the bag, not running	200 — collected 3, starting 2	56
43	the stick goes back in	banner gone	
44	not running, before collecting	409 — not_checked_in	
45	the live table on the wall new	3 of 12 bags handed over; Server running	31
46	the Wi-Fi drops: a check-in	saved — 1 check-in waiting to send. Show queued scans	51
47	the Wi-Fi comes back	200 — sent; recorded at 16:07 (the phone's time)	
48	a phone that died with a scan on it: recovered off its screen	200 — Mevrou Steyn	54
49	Export all on Friday night new	server stopped; files written	57
50	the wall while the server is down new	Server not running	57
51	Restart server after the export new	200 — resumed with 5 collected	
52	Export this class: LTR 65, with the desk open new	server running again after 0.6 s	59
53	type a race number off the slip	200 — 2 bags, next 1020	42
54	a bag that is not on that slip	409 — wrong_group	44
55	bag only, the second bag on the slip	200 — bypass recorded	61
56	one code opens the whole registration	200 — 3 bags: 512, 1016, 1017	8
57	scan bag BAG-21700202	200 — 2 left	62
58	scan bag BAG-21700200	200 — 1 left	
59	scan bag BAG-21700201	200 — 0 left	63
60	the same runner a second time	200 — already collected	65
61	a bag printed for somebody else	409 — wrong_bag	67
62	a code from another race	404 — No entry with that ID	68
63	the table app opens	200 — mode audit	70

#	what was tried	answer	figure
64	a bag nobody collected	200 — no conflict	71
65	a bag the record says has gone	200 — conflict flagged	72
66	the audit report	200 — 2 found, 1 conflict, 1 unaccounted	73
67	finish the fallback while scanning works	409 — not_fallback	
68	the desk finds a registration collected by someone else (option 3) new	200 — Radebe: delegate Gerda van Wyk (No 540)	74
69	the collector hands in the pre-printed slip: Palesa's bag new	200 — bypass recorded	75
70	a runner who paid at the desk and ran: on the start list? new	3 on the LTR 65 list, Venter named as paid at the desk	
71	the desk declares scanning has stopped	200 — mode fallback	23
72	everyone else collected their bag	200 — 0 marked collected, 1 left	78
73	a PC started with --pin, a phone without one new	401 — pin	79
74	...with the PIN typed in new	200 — scanning	
75	labels: 2 x 6 on A4 with the default geometry new	102 x 47.2 mm, border 2/2.0 mm, top 2, bottom 2.0, gaps 2	3

All of them behaved as this book says they do. Two are worth reading twice. The one where the fallback is finished while scanning is working: the app that finishes the paper fallback refuses to do so, because the difference between an audit and a record is the most expensive mistake in the system and it must not be reachable by accident. And the one where the phone is panned past another runner's bag on its way to the right one: nothing was sent to the laptop at all.

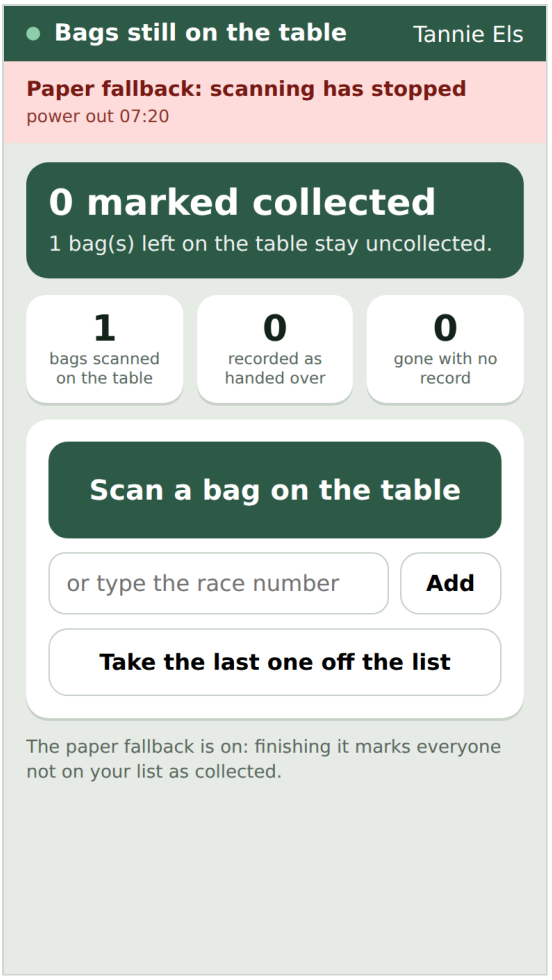


Figure 78. *After the desk had declared that scanning stopped and the table was walked again: everyone whose bag was not on the walker's list is recorded as having collected it — by then every one of them already had — and the count of what is really still on the table has not moved.*

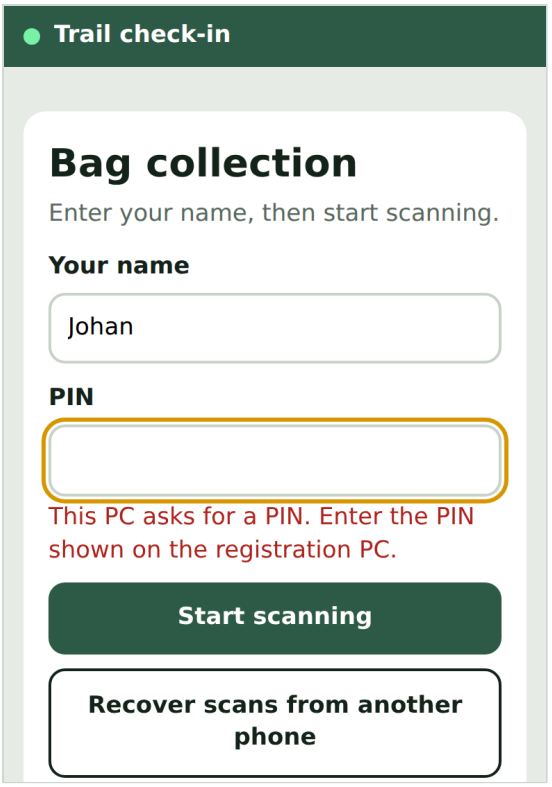


Figure 79. *A laptop started with a PIN, and a phone that was never told about one. The field appears only now, and only on that phone.*

And what the server itself printed while all of that was happening, which is the same log the desk watches on the day:

```
Loading entries from input/race_day.xlsx
SLIP TORN OFF for Mthembu (2 bag(s)), registration 21700110
PAPER FALLBACK ON: drill: power out at the marquee
scanning declared working again: drill: power out at the marquee
CHECK-IN 21700203 Sibusiso Ngcobo bag=BAG-21700203 -> checkins_00002_20261001-160546.xlsx
Loading entries from input/race_day.xlsx
output/checkin_files was missing: created again
CHECK-IN 21700203 Sibusiso Ngcobo bag=BAG-21700203 -> checkins_00002_20261001-160549.xlsx
CHECK-IN 21700203 Sibusiso Ngcobo bag=BAG-21700203 -> checkins_00004_20261001-160601.xlsx
CHECK-IN 21700204 Jurie Oosthuizen bag=BAG-21700204 -> checkins_00005_20261001-160628.xlsx
CHECK-IN 21700210 Marius Venter bag=BAG-21700210 -> checkins_00006_20261001-160642.xlsx
NOT RUNNING: 31 Marius Venter  (left out of the start list)
CHECK-IN 21700205 Lerato Fourie bag=BAG-21700205 -> checkins_00008_20261001-160727.xlsx
CHECK-IN 21700212 Mevrou Steyn bag=BAG-21700212 -> checkins_00009_20261001-160731.xlsx
Resuming from state.json (version 9)
Resuming from state.json (version 9)
SLIP TORN OFF for Mthembu (2 bag(s)), registration 21700110
CHECK-IN 21700206 Anna Mthembu bag=BAG-21700206 -> checkins_00010_20261001-160749.xlsx
CHECK-IN 21700207 Pieter Mthembu bag=BAG-21700207 -> checkins_00011_20261001-160756.xlsx
CHECK-IN 21700202 Ruben Kruger bag=BAG-21700202 -> checkins_00012_20261001-160800.xlsx
CHECK-IN 21700200 Werner Botha bag=BAG-21700200 -> checkins_00013_20261001-160804.xlsx
CHECK-IN 21700201 Ruan Kruger bag=BAG-21700201 -> checkins_00014_20261001-160808.xlsx
CHECK-IN 21700208 Gerda van Wyk bag=BAG-21700208 -> checkins_00015_20261001-160818.xlsx
ON THE TABLE 21700211 Christo Nel
ON THE TABLE 31 Marius Venter  ** RECORDED AS HANDED OVER **
SLIP TORN OFF for Radebe (1 bag(s)), registration 21700112
CHECK-IN 21700209 Palesa Radebe bag=BAG-21700209 -> checkins_00016_20261001-160833.xlsx
Resuming from state.json (version 16)
PAPER FALLBACK ON: power out 07:20
ON THE TABLE 21700211 Christo Nel
```